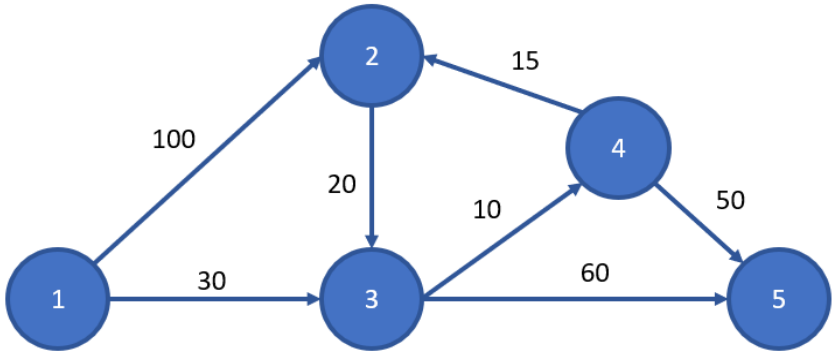
Networks

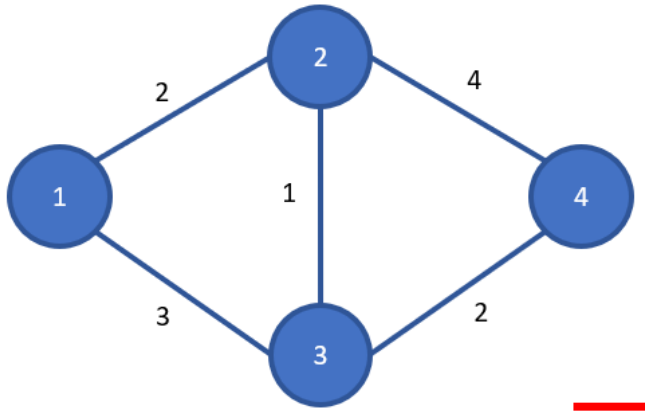
Shortest Paths and Minimum Spanning Tree (MST)

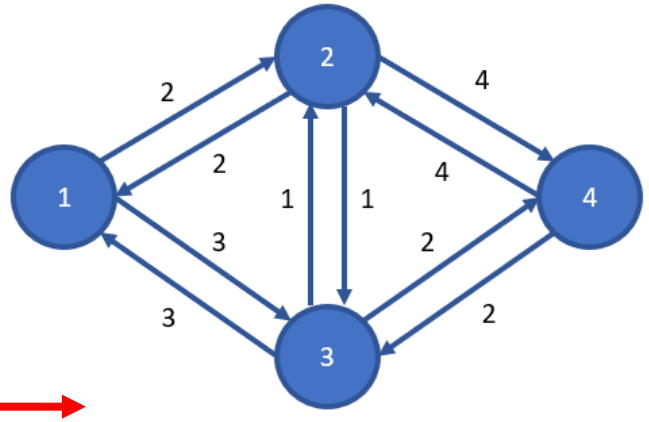
Sensitivity: Internal

Network

Directed graph $G = (V, E)$, where E is a set of edges, subset of $\mathrm{V} \times \mathrm{V}$, and a weight function w: $E \to \mathbb{R}$







$V = \{1,2,3,4,5\}$
$\mathrm{E} = \{(1,2), (1,3), (2,3), (3,4), (3,5), (4,2), (4,5)\}$

e.g. $w(1,2) = 100$

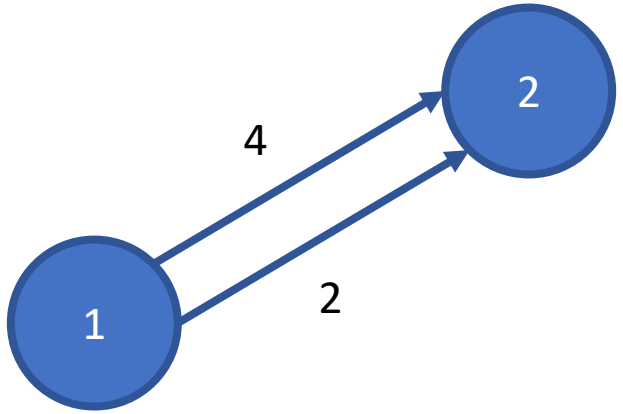
$V = \{1,2,3,4\}$
$E = \{(1,2), (1,3), (2,3), (2,4), (3,4)\}$

$V = \{1,2,3,4\}$
$E = \{(1,2), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,2), (4,3)\}$

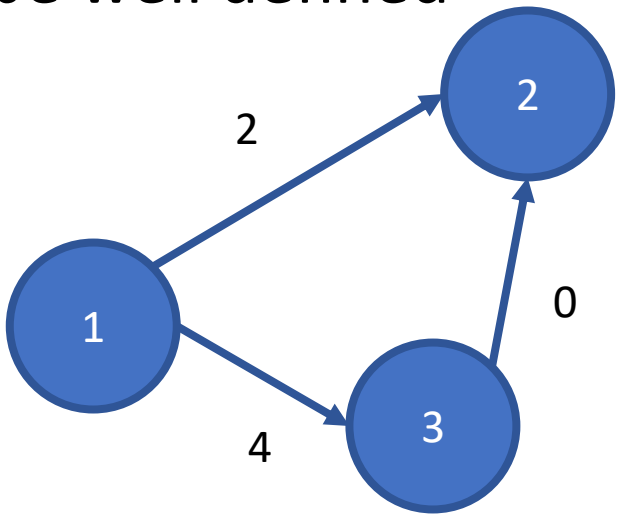
Sensitivity: Internal

Network

Note we do not allow multiple edges between vertices u and v, otherwise the edge weight would not be well defined



What is $w(1,2)$?



What is $w(1,2)$?

$$w(1,2) = 2$$

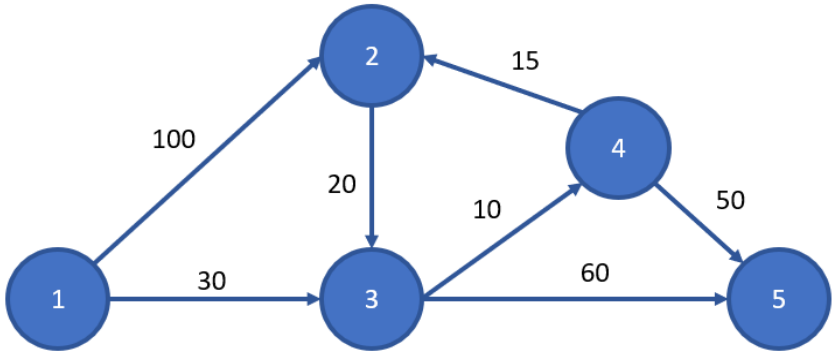
Alternative path from 1-2 is 1-3-2

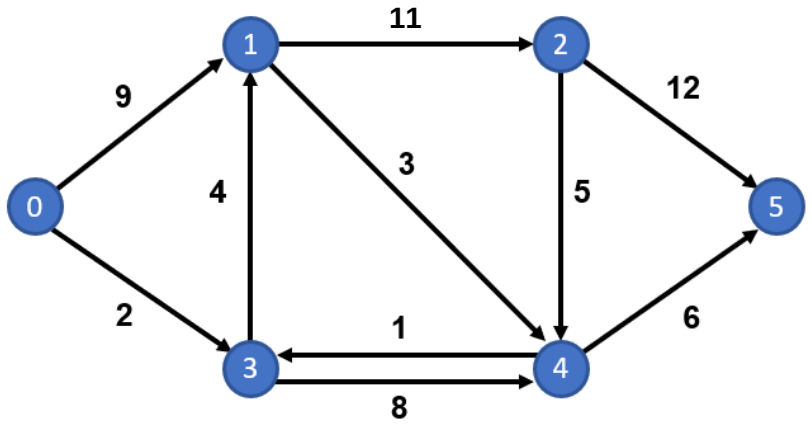
$$w(1,3) + w(3,2) = 4 + 0 = 4$$

Sensitivity: Internal

Adjacency Matrix

We won't have direct use for this, but it is worth noting!





$$\begin{pmatrix} 0 & 100 & 30 & - & - \\ - & 0 & 20 & - & - \\ - & - & 0 & 10 & 60 \\ - & 15 & - & 0 & 50 \\ - & - & - & - & 0 \end{pmatrix} \textbf{ OR } \begin{pmatrix} 0 & 100 & 30 & \infty & \infty \\ \infty & 0 & 20 & \infty & \infty \\ \infty & \infty & 0 & 10 & 60 \\ \infty & 15 & \infty & 0 & 50 \\ \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 9 & - & 2 & - & - \\ - & 0 & 11 & - & 3 & - \\ - & - & 0 & - & 5 & 12 \\ - & 4 & - & 0 & 8 & - \\ - & - & - & 1 & 0 & 6 \\ - & - & - & - & - & 0 \end{pmatrix} \textbf{ OR } \begin{pmatrix} 0 & 9 & \infty & 2 & \infty & \infty \\ \infty & 0 & 11 & \infty & 3 & \infty \\ \infty & \infty & 0 & \infty & 5 & 12 \\ \infty & 4 & \infty & 0 & 8 & \infty \\ \infty & \infty & \infty & 1 & 0 & 6 \\ \infty & \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

Sensitivity: Internal

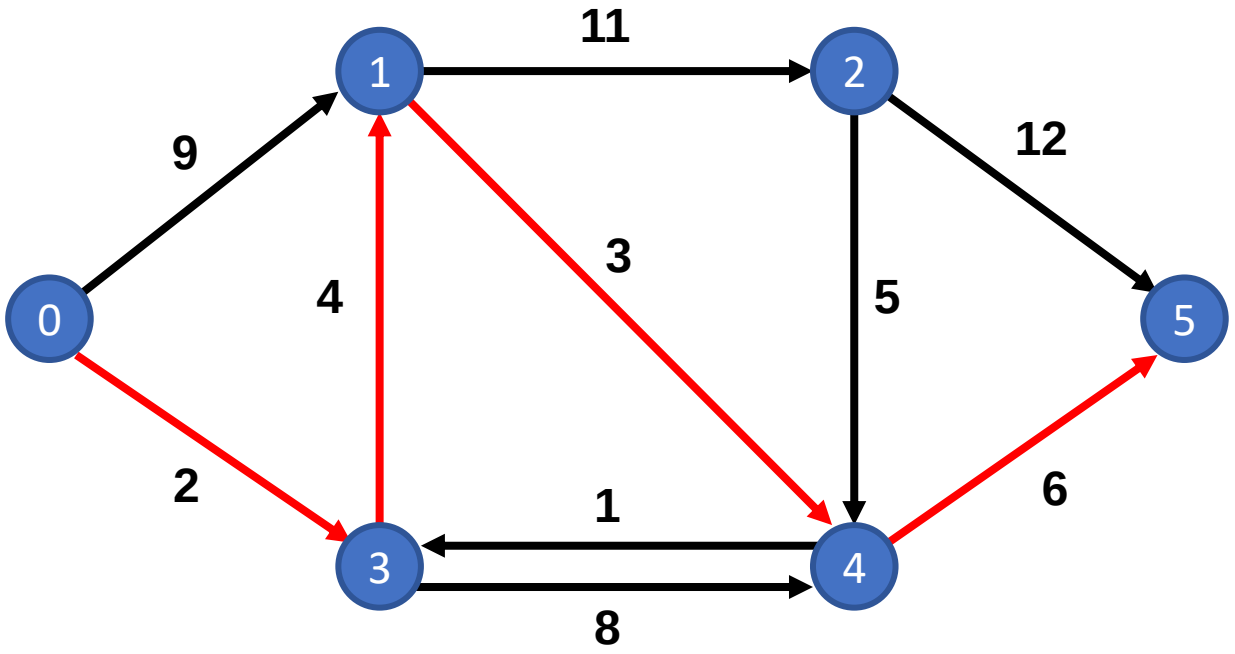
Shortest Path Problem

Definition: Find the shortest path between two vertices (source and destination) such that the sum of the weights of its edges is minimised.

- Perhaps one of the most fundamental problems in Mathematics and Computer Science
- Huge implications on routing and traffic problems, e.g. vehicular, telecommunications.
- If you've used Google Maps, you've asked to solve a form of shortest path problem!

Sensitivity: Internal

Blue Peter Example



Shortest path between 0 and 5 is 0-3-1-4-5 of length 15

Sensitivity: Internal

Dijkstra's Algorithm

- Designed by Edsger Wybe Dijkstra in 1956 (published 1959) to demonstrate the new ARMAC computer
- Still widely used and researched today!
- Route planners use pre-processing speed-up techniques such as contraction hierarchies.

Edge weights need to be nonnegative! Otherwise use Bellman-Ford

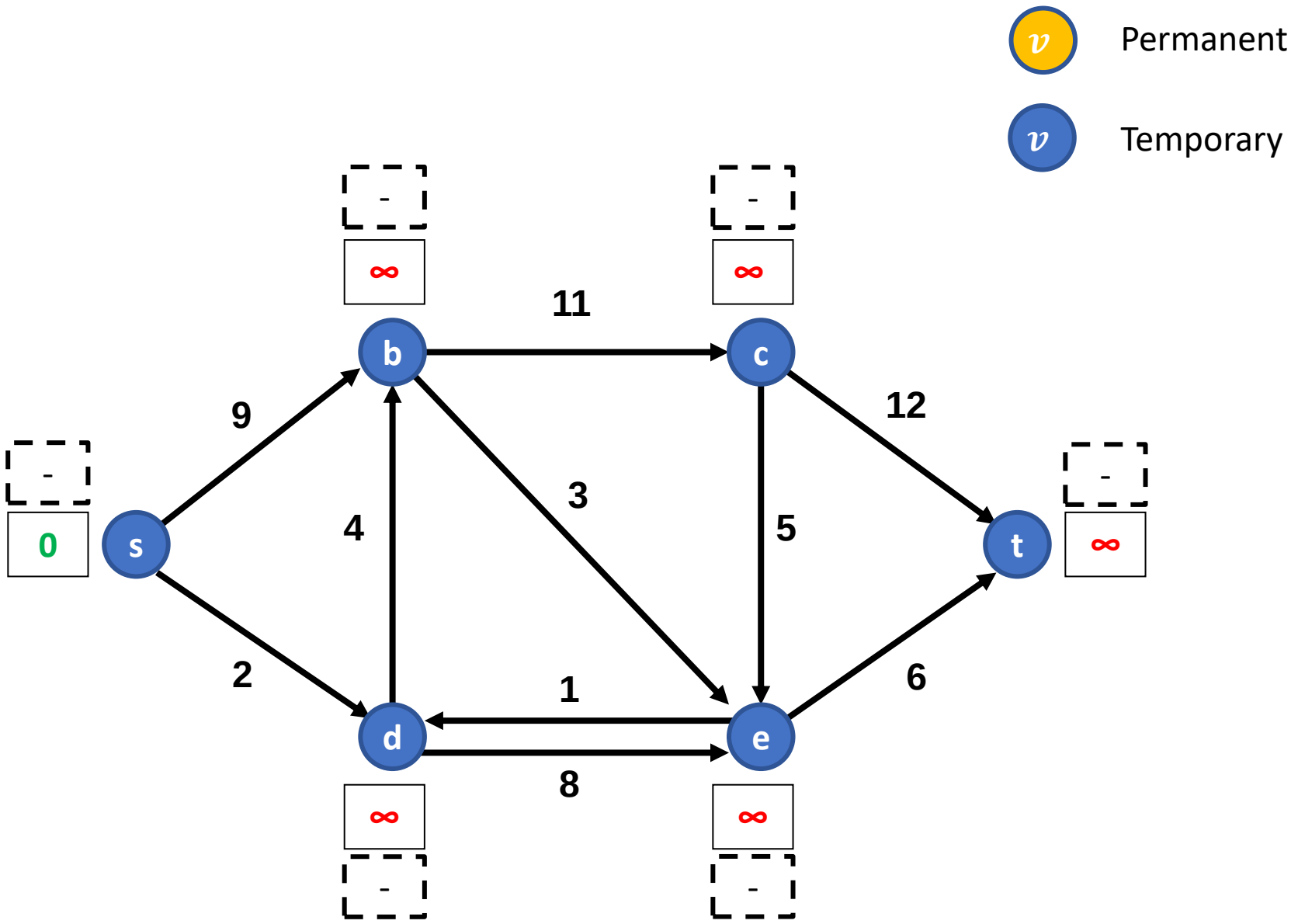
Homework: Investigate Bellman-Ford Algorithm, this can also detect negative weight cycles

1. Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞, label all vertices as **temporary**
2. Label the vertex with minimum distance as **permanent**
3. Update the distances and predecessors to reachable **temporary** vertices from the current permanent vertex
4. Repeat steps 2 and 3 until all vertices are marked **permanent**

Sensitivity: Internal

Set Up

v	Label	d_v	π_v
s	T	0	-
b	T		
c	T		
d	T		
e	T		
t	T		



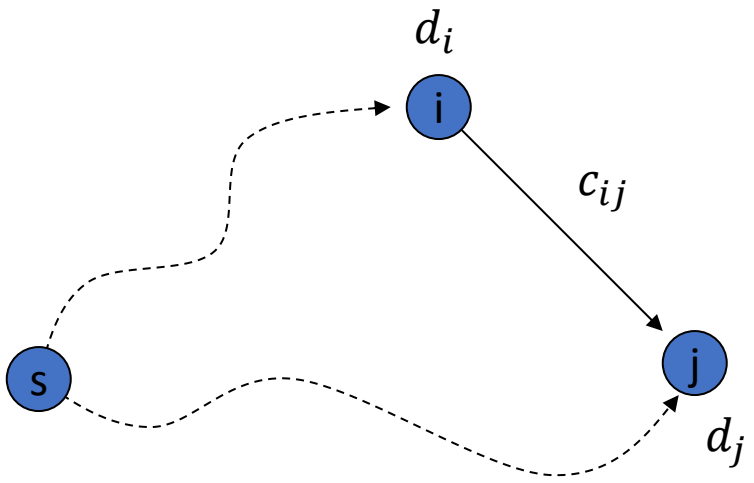
Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞

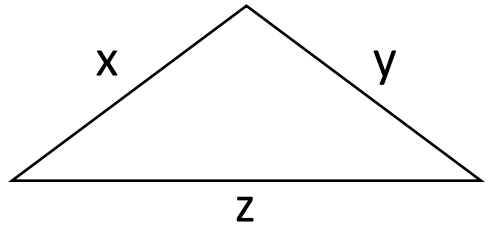
Sensitivity: Internal

Improving Cost Labels

- Shortest path costs satisfy the triangle inequality:

  $d_i + c_{ij} \geq d_j$ for every edge (i, j).





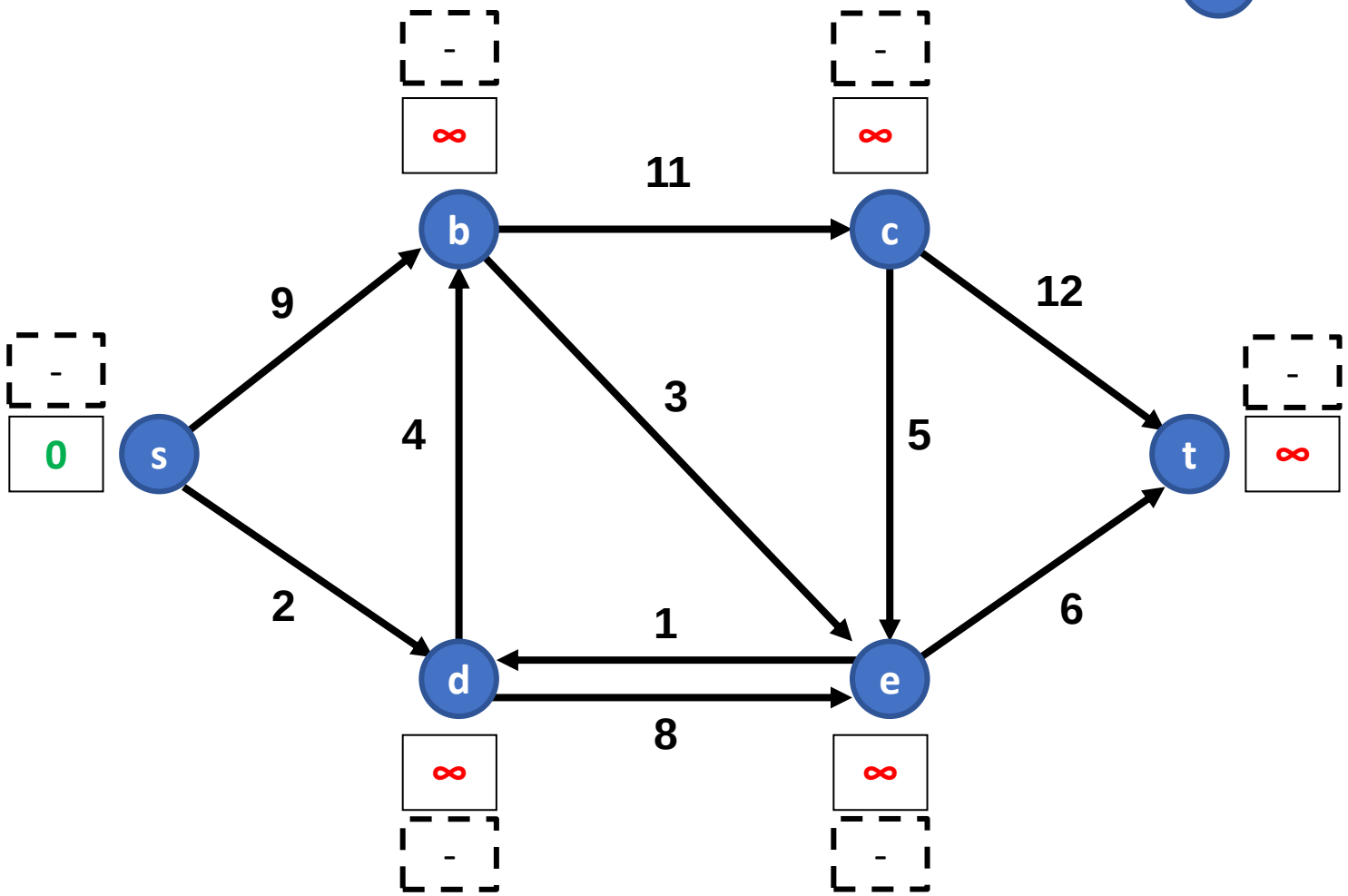
x + y ≥ z

- If this is not **TRUE** then there exists some path that improves d_j
  - E.g. The path to d_i + the edge (i, j)

Sensitivity: Internal

Dijkstra Example

v	Label	d_v	π_v
s	T	0	-
b	T		
c	T		
d	T		
e	T		
t	T		

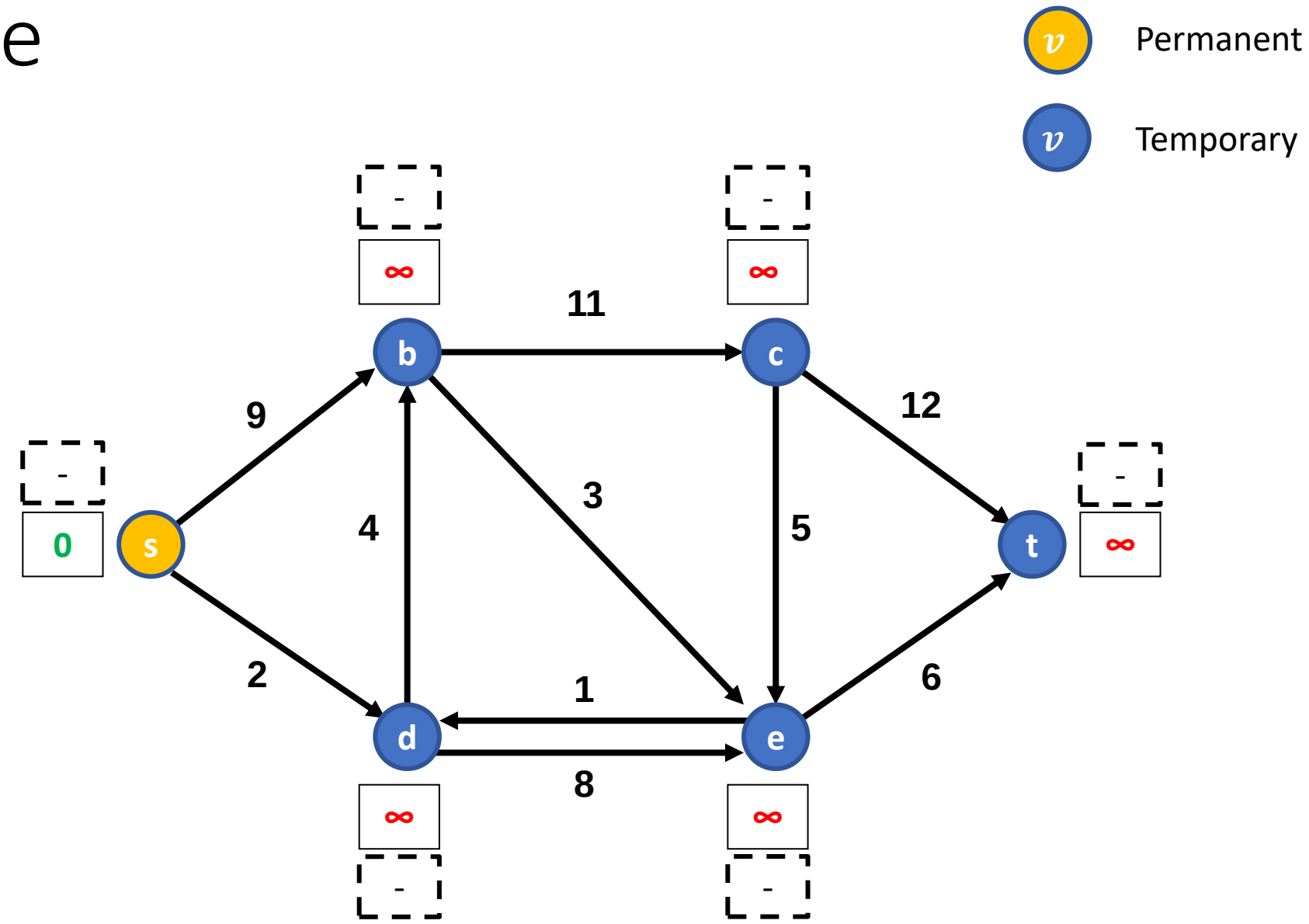


Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	T		
c	T		
d	T		
e	T		
t	T		

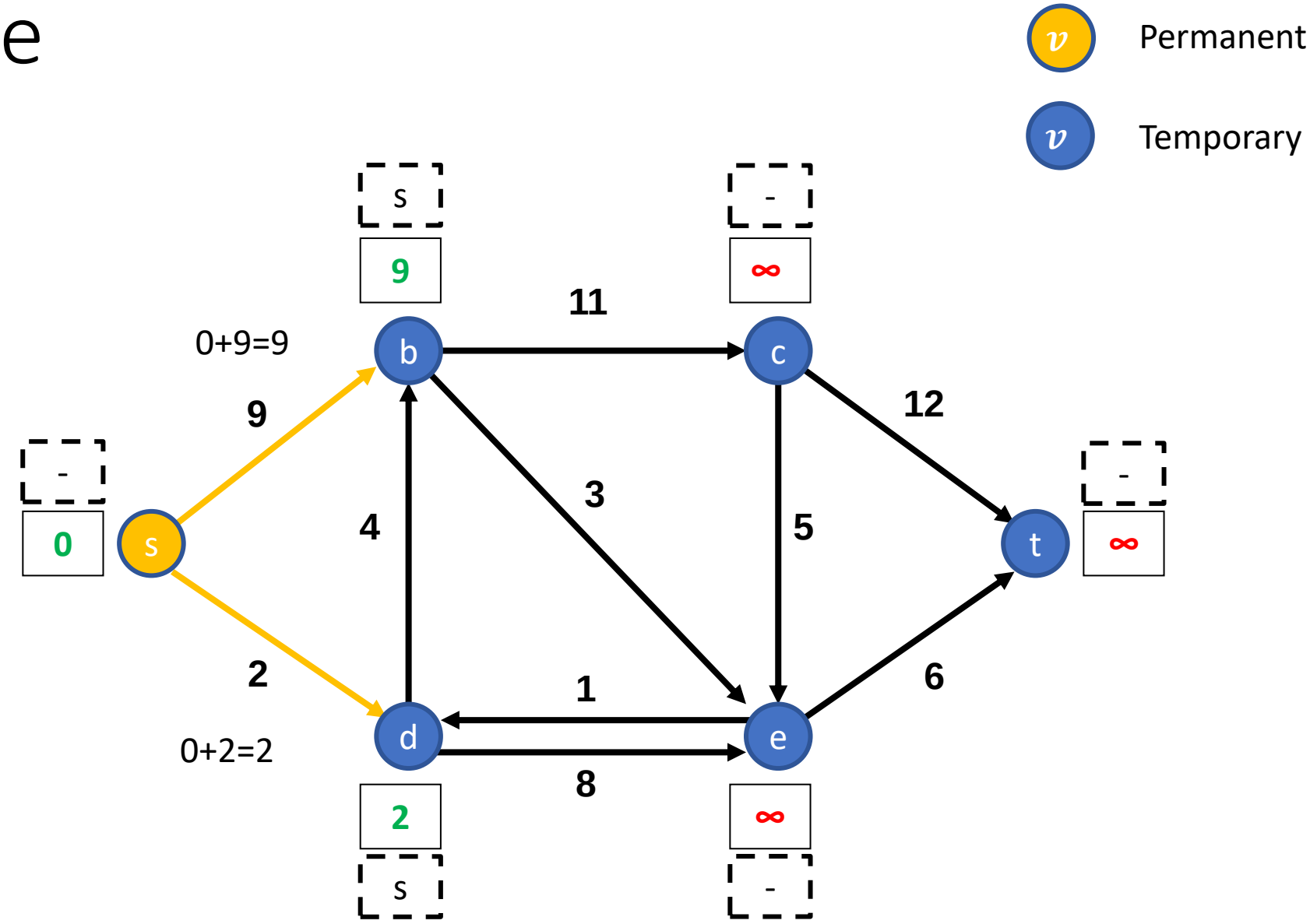


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	T	9	s
c	T		
d	T	2	s
e	T		
t	T		

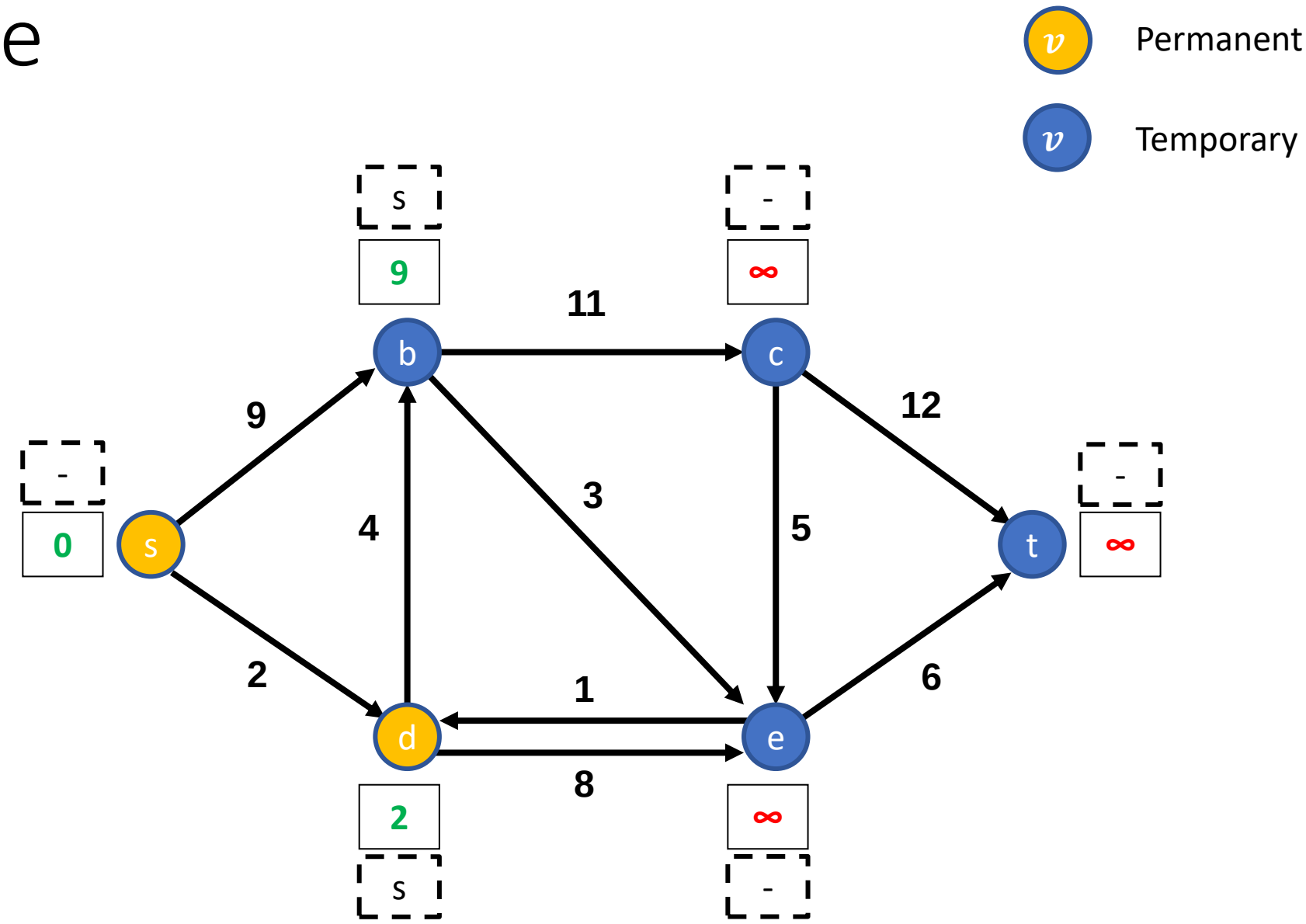


Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	T	9	s
c	T		
d	P	2	s
e	T		
t	T		

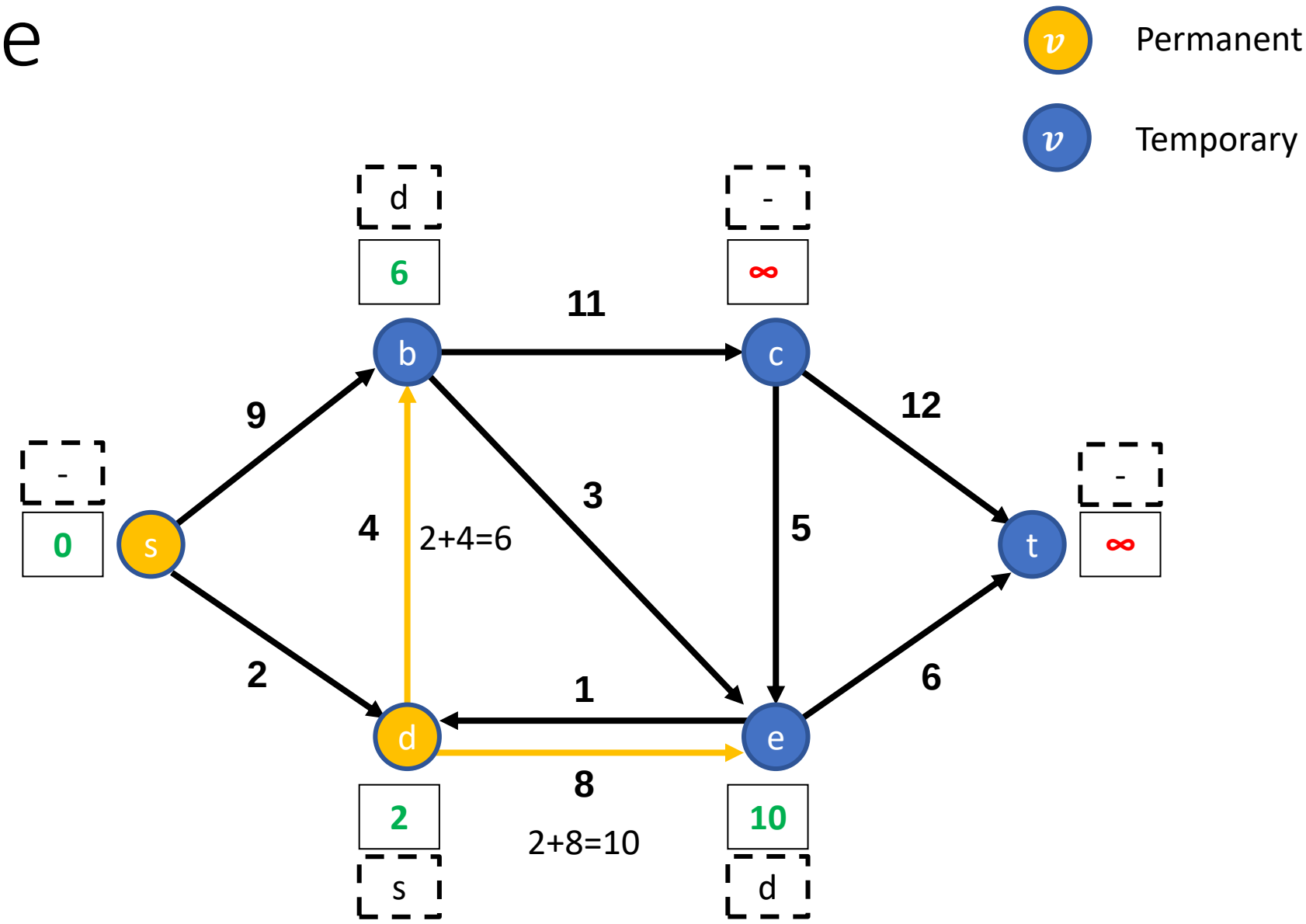


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	T	~~9~~ 6	~~s~~ d
c	T		
d	P	2	s
e	T	10	d
t	T		

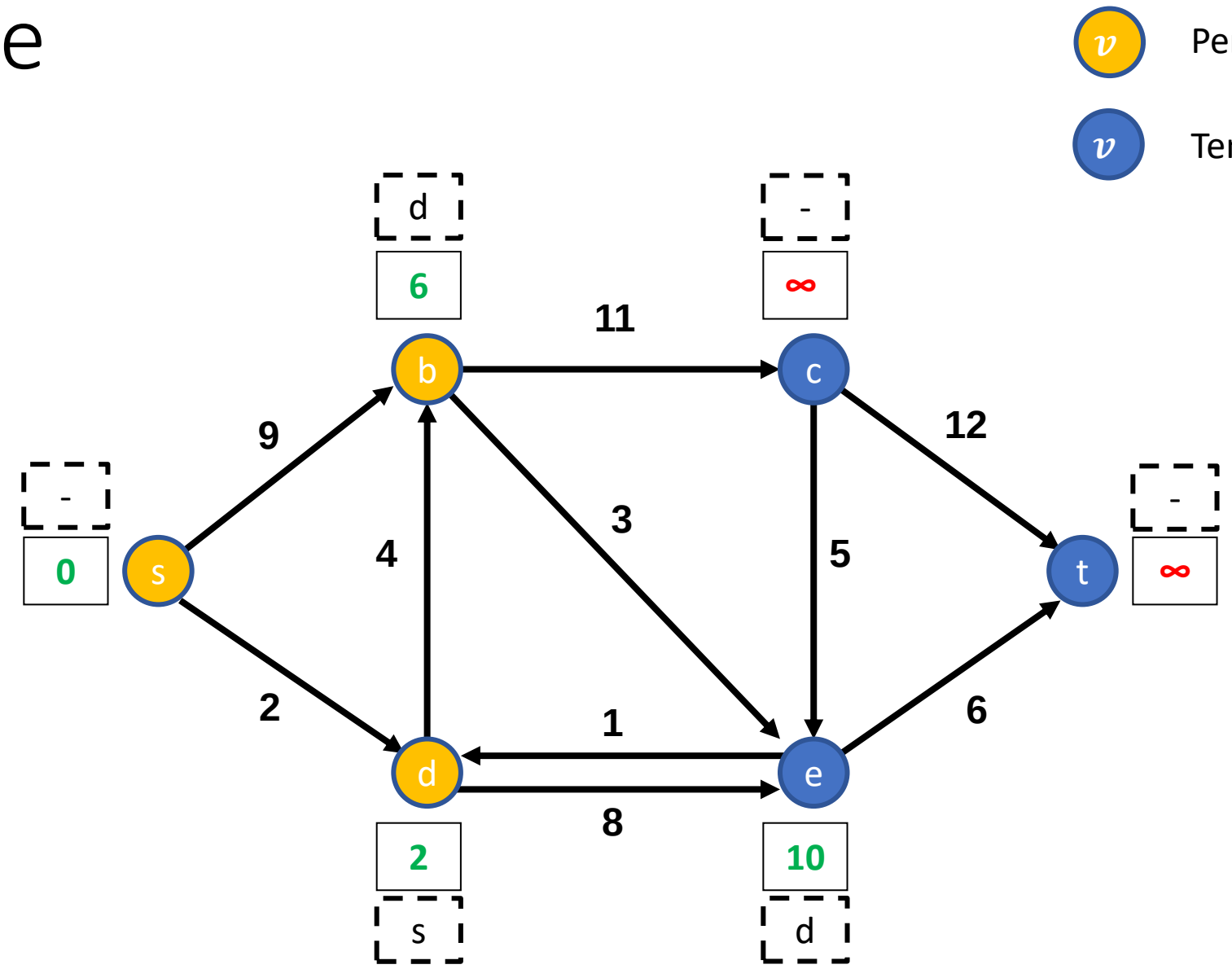


Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T		
d	P	2	s
e	T	10	d
t	T		

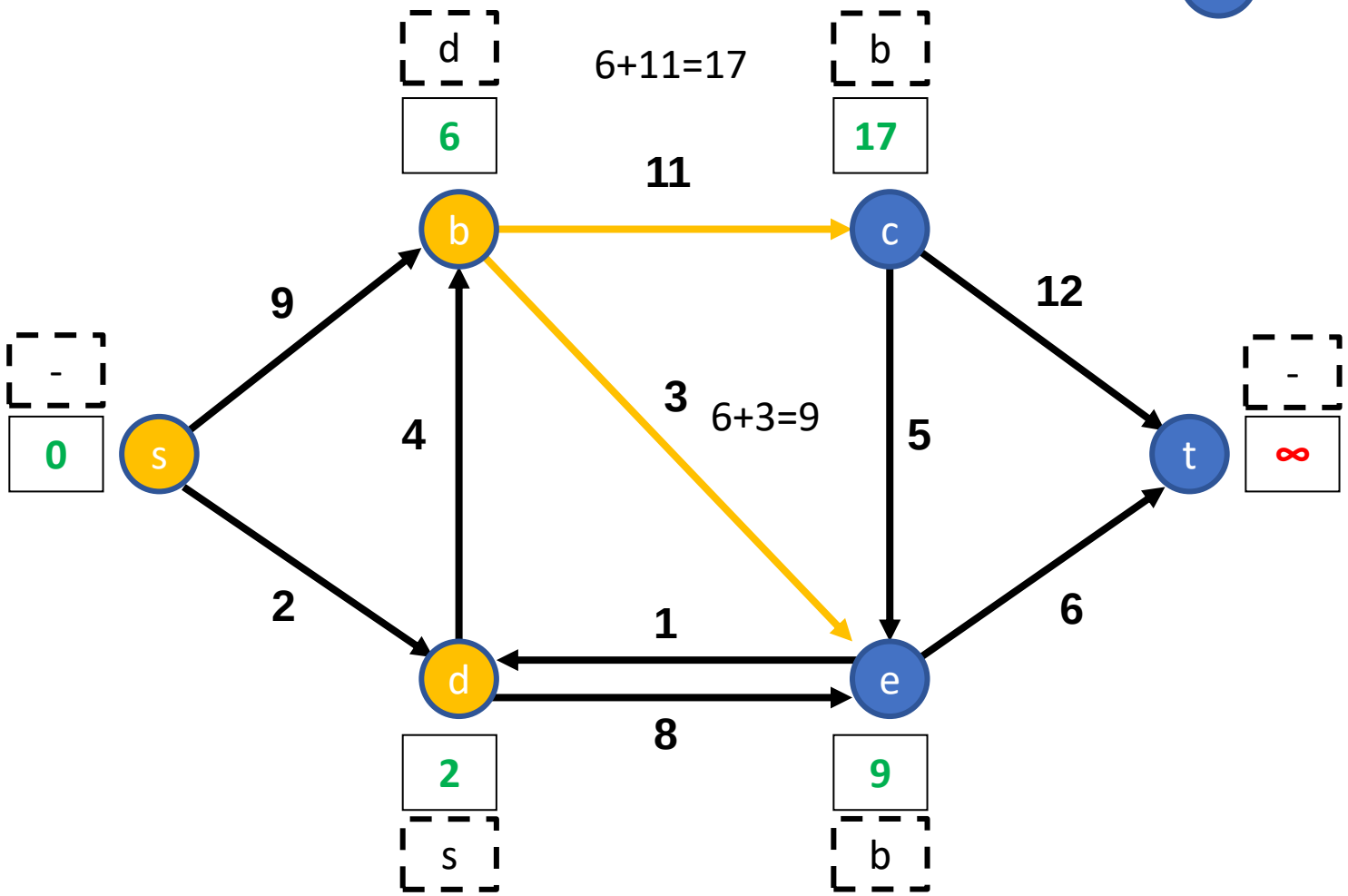


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T	17	b
d	P	2	s
e	T	~~10~~ 9	~~d~~ b
t	T		

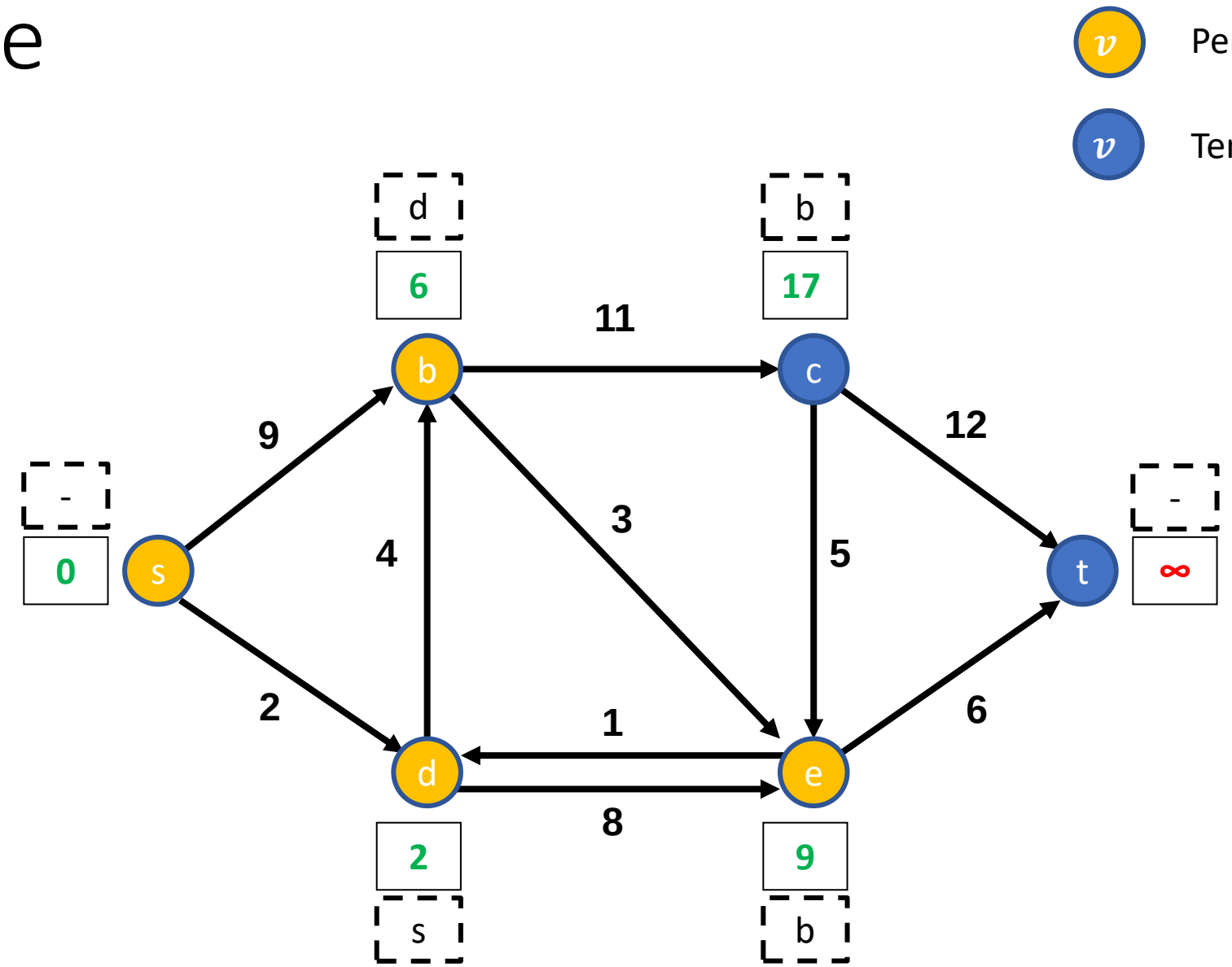


Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	T		

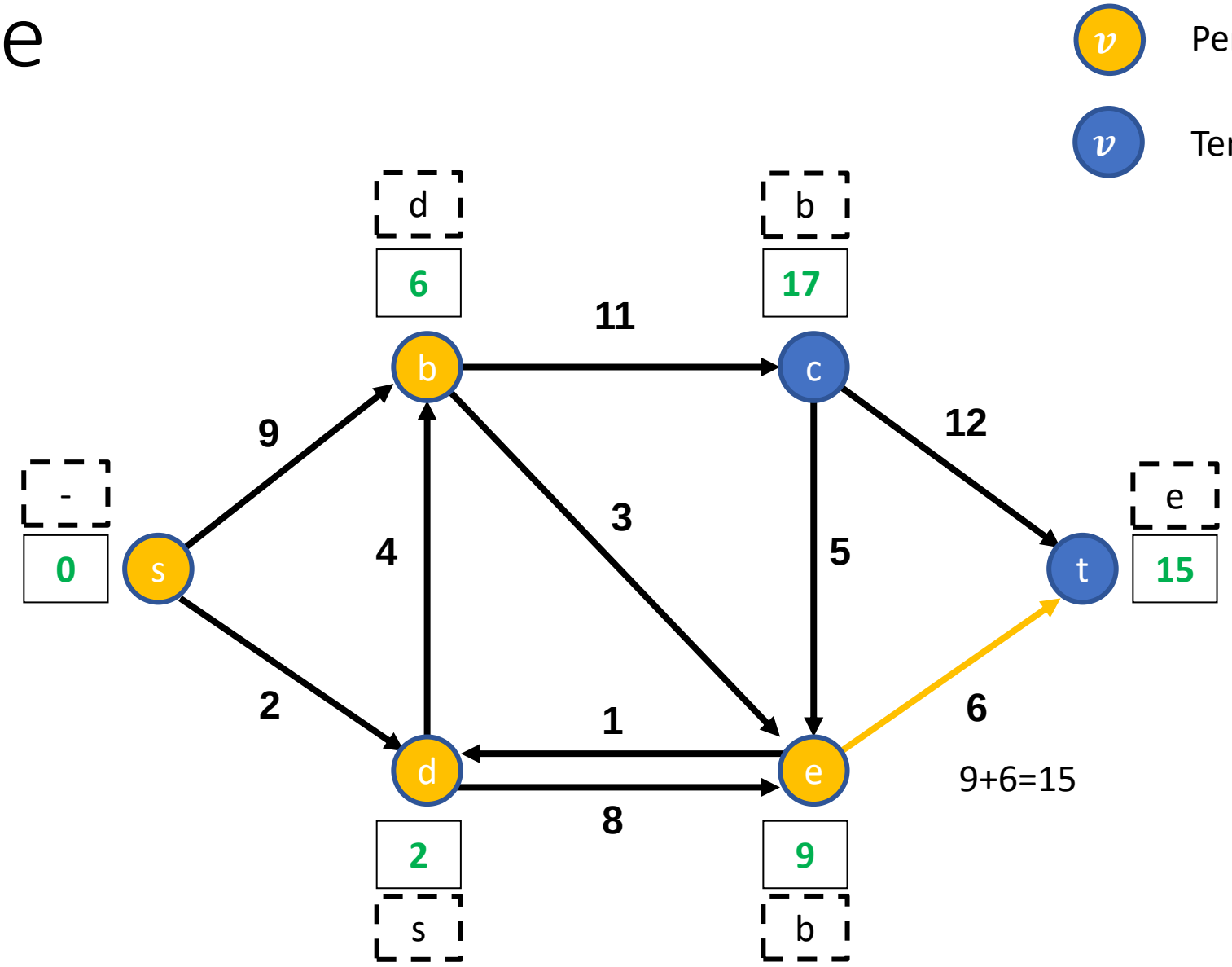


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	T	15	e

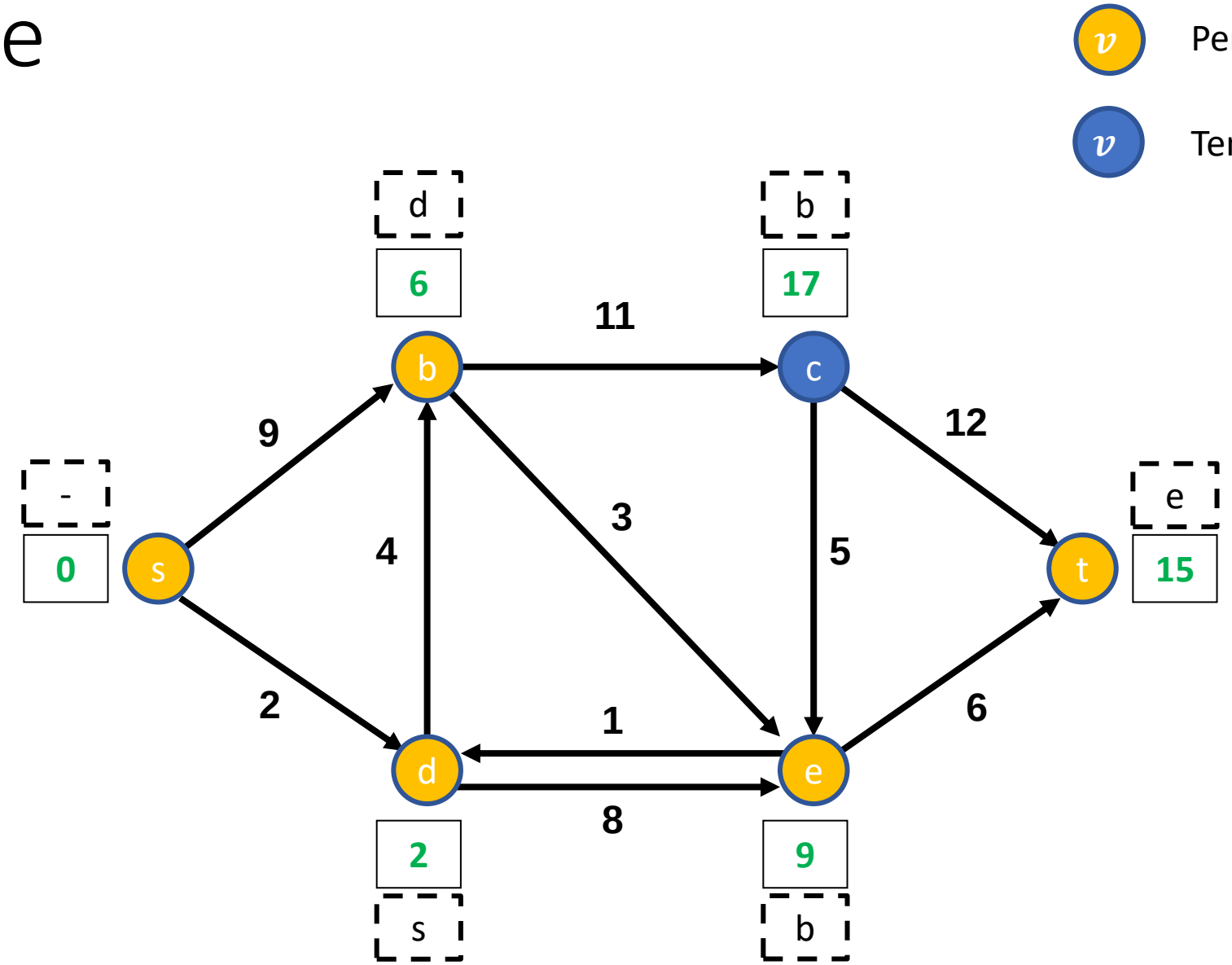


Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

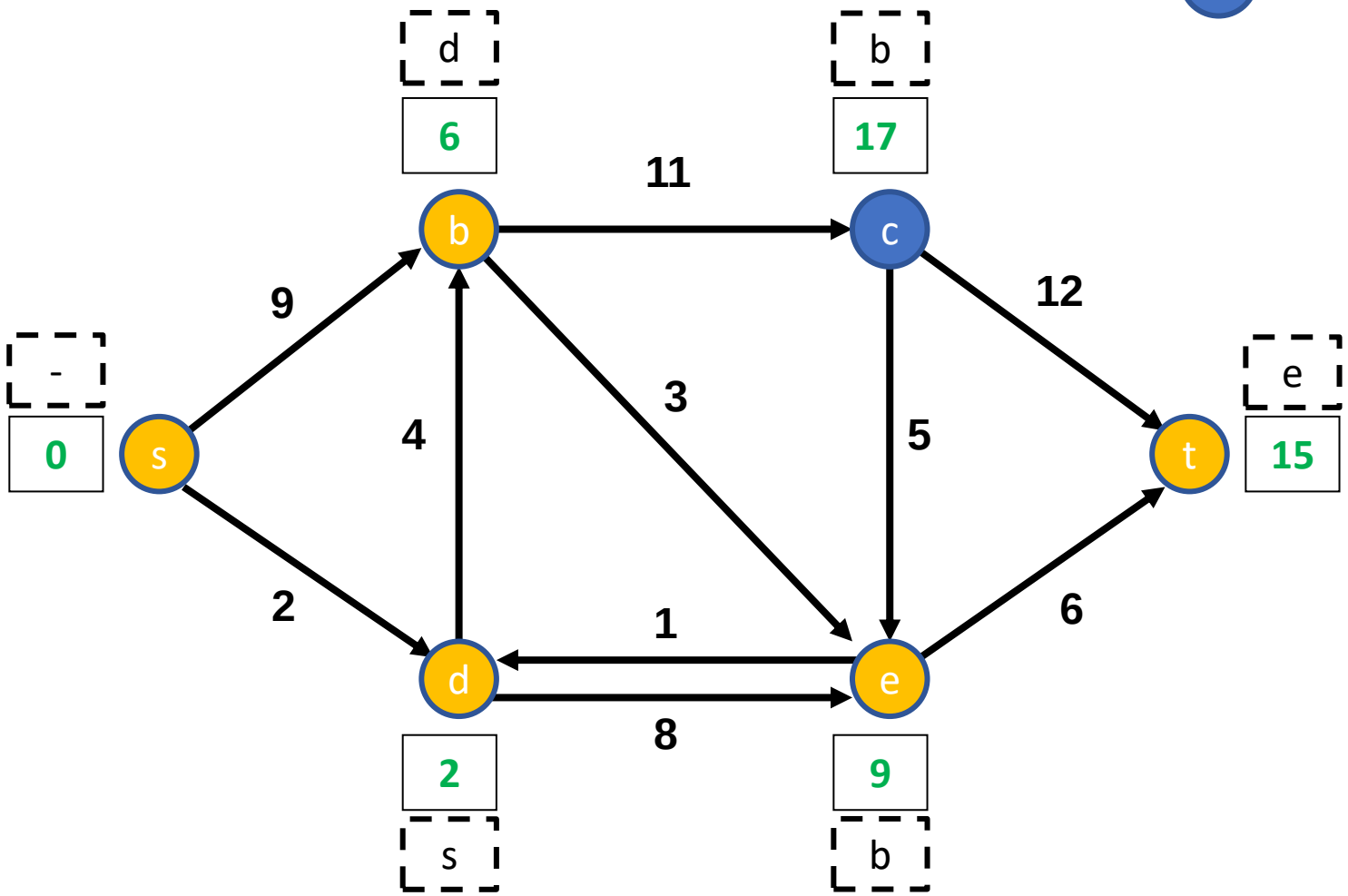


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	T	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

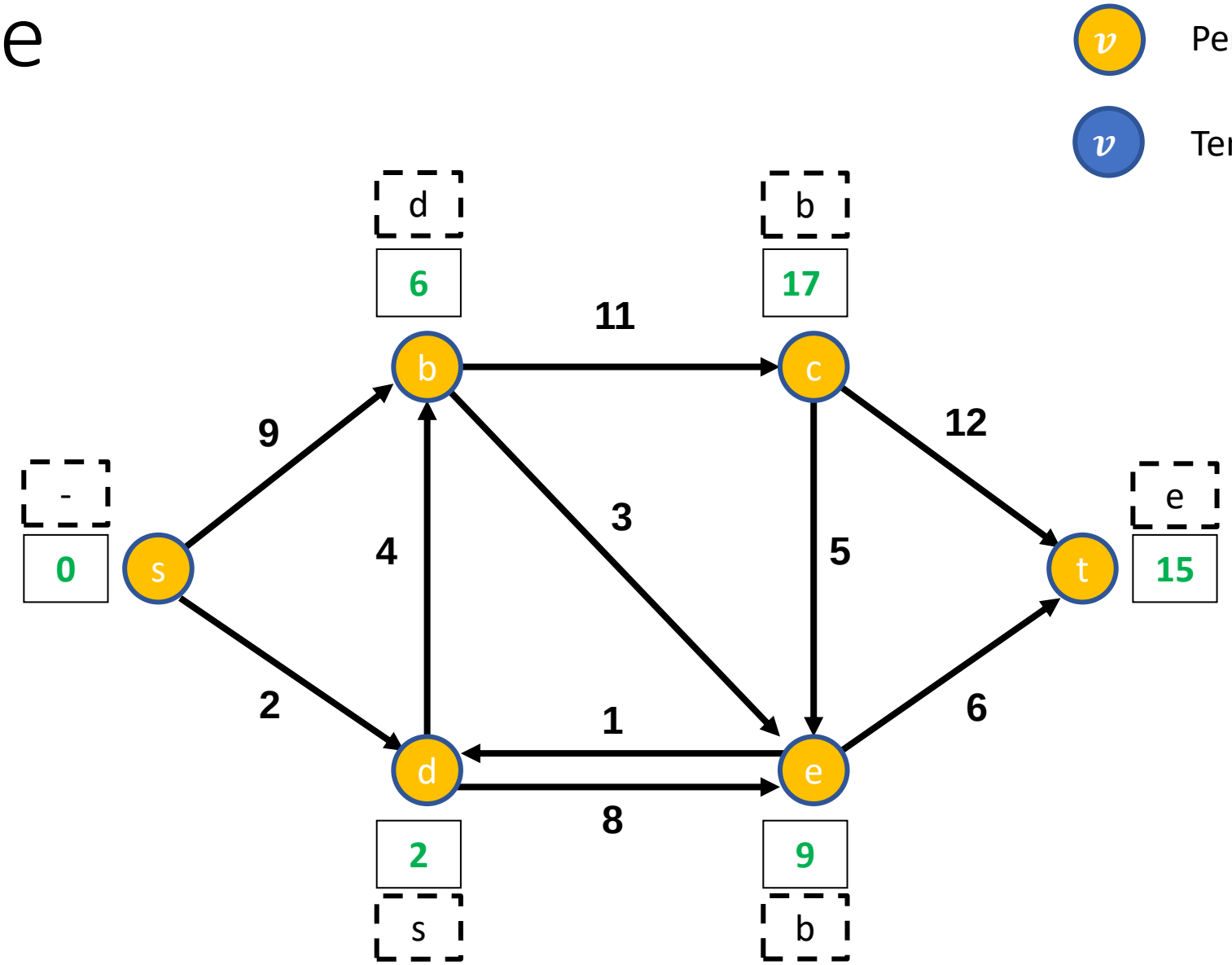


Update distances to reachable vertices which are labelled temporary. No changes

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

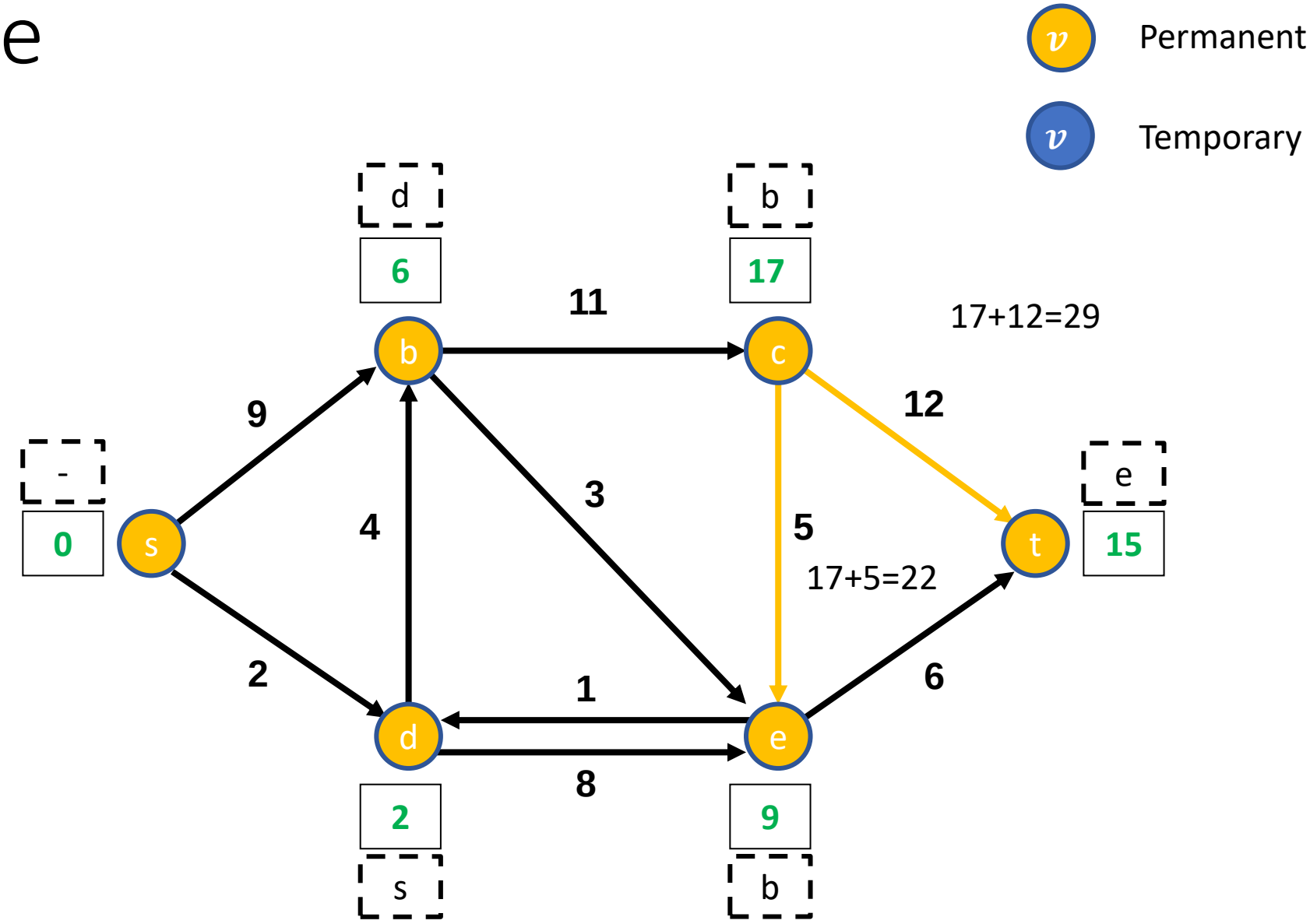


Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

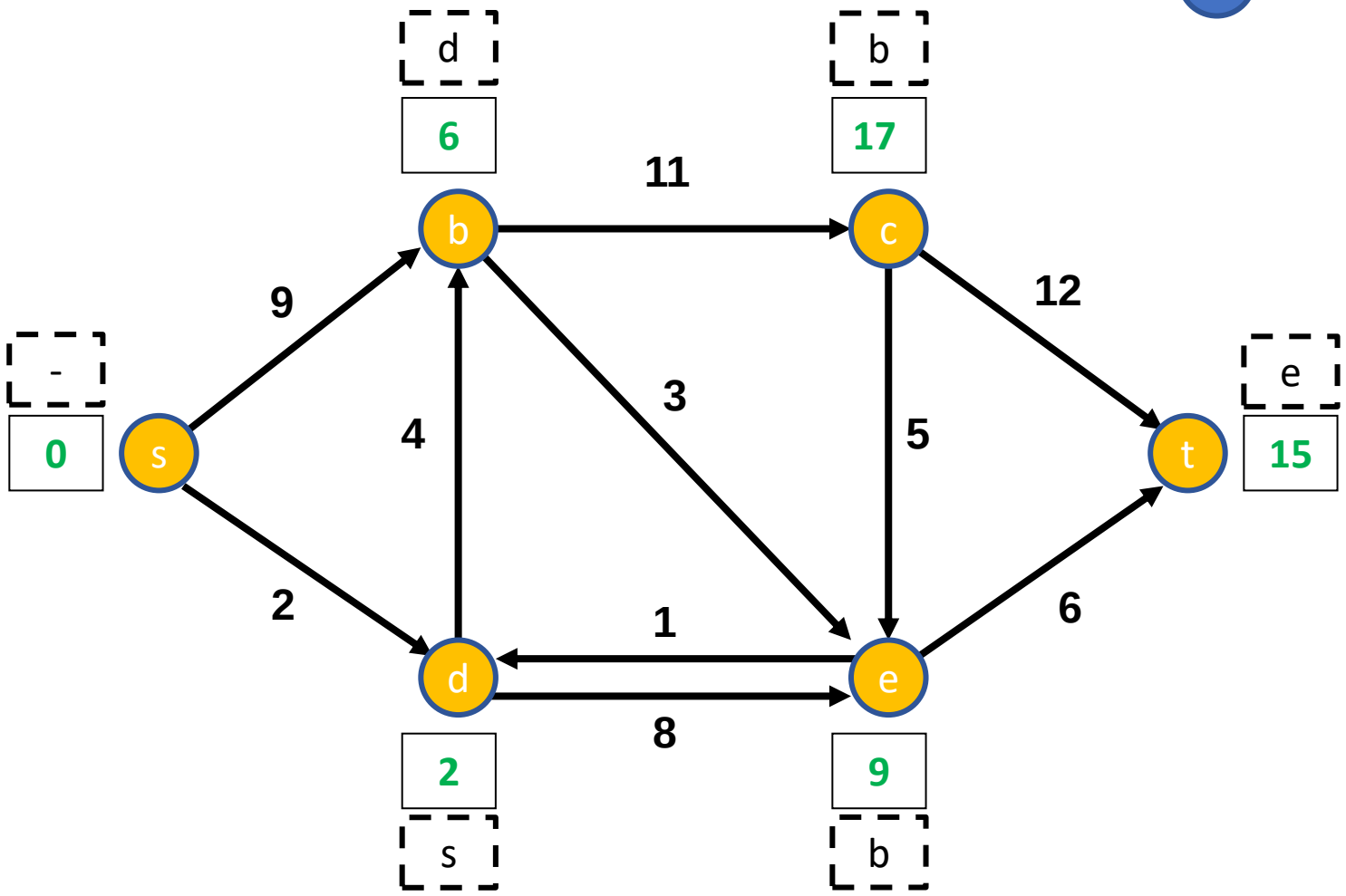


Update distances to reachable vertices which are labelled temporary. No changes

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

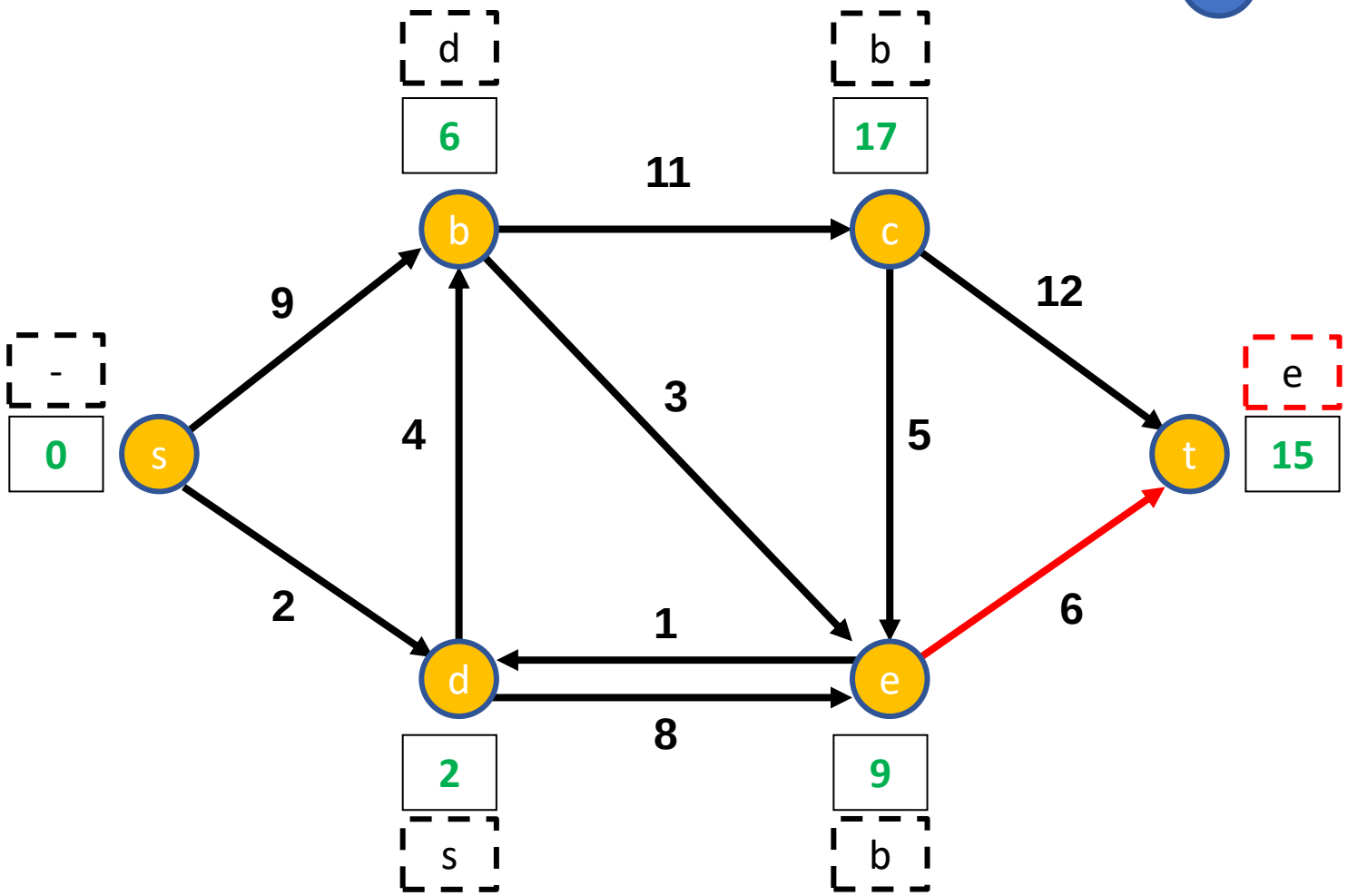


Final Network and Table

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

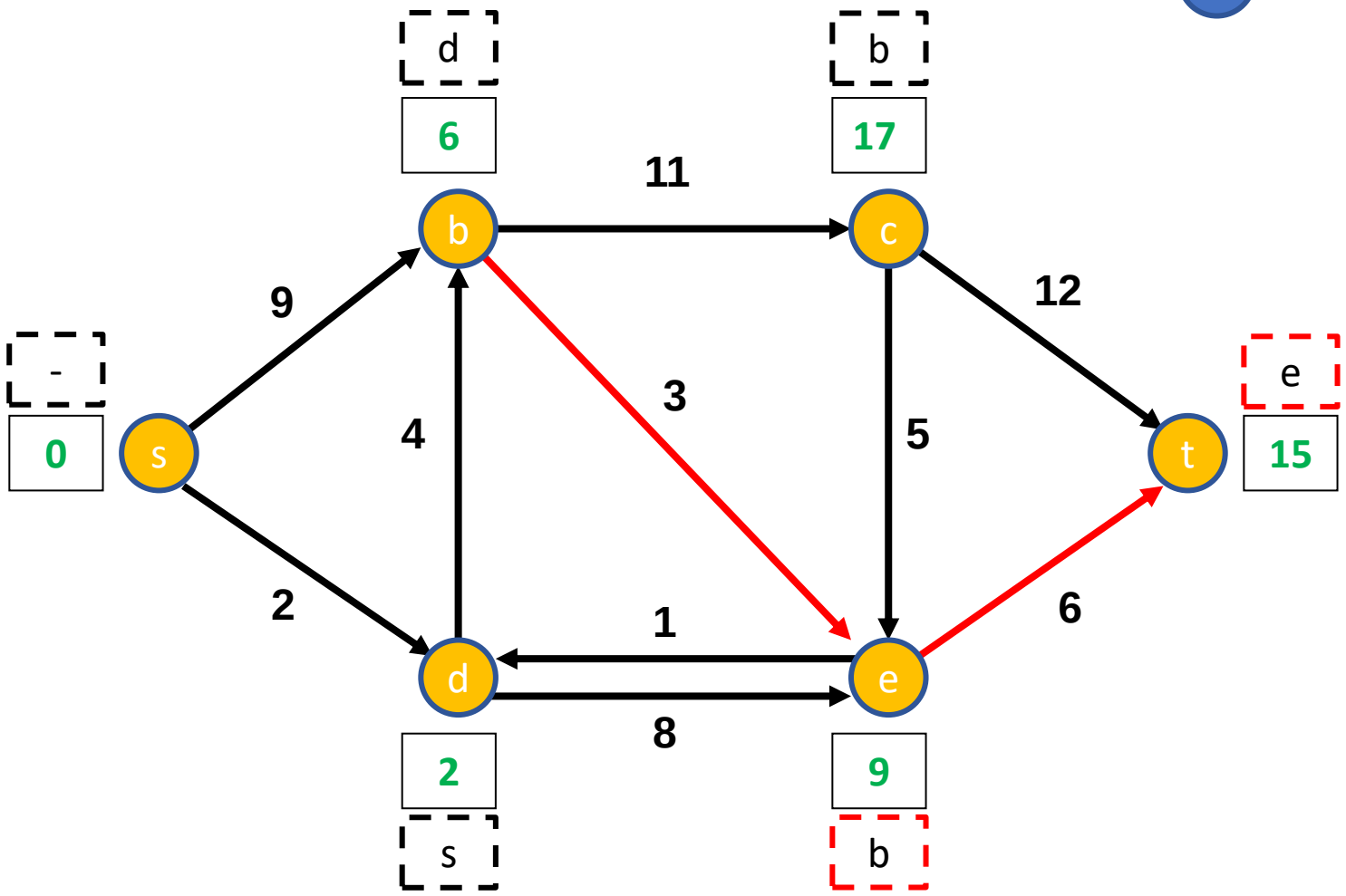


To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

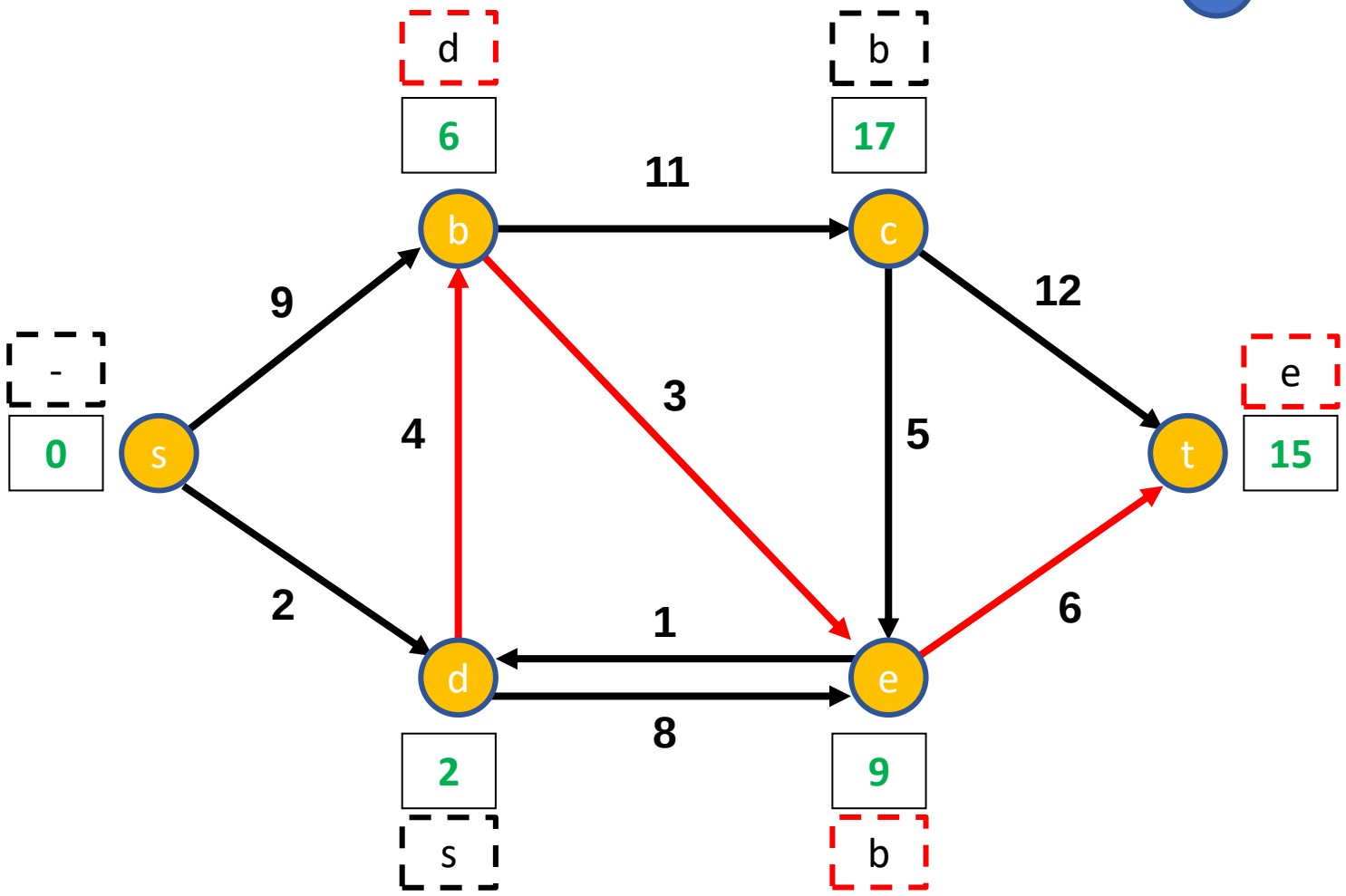


To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

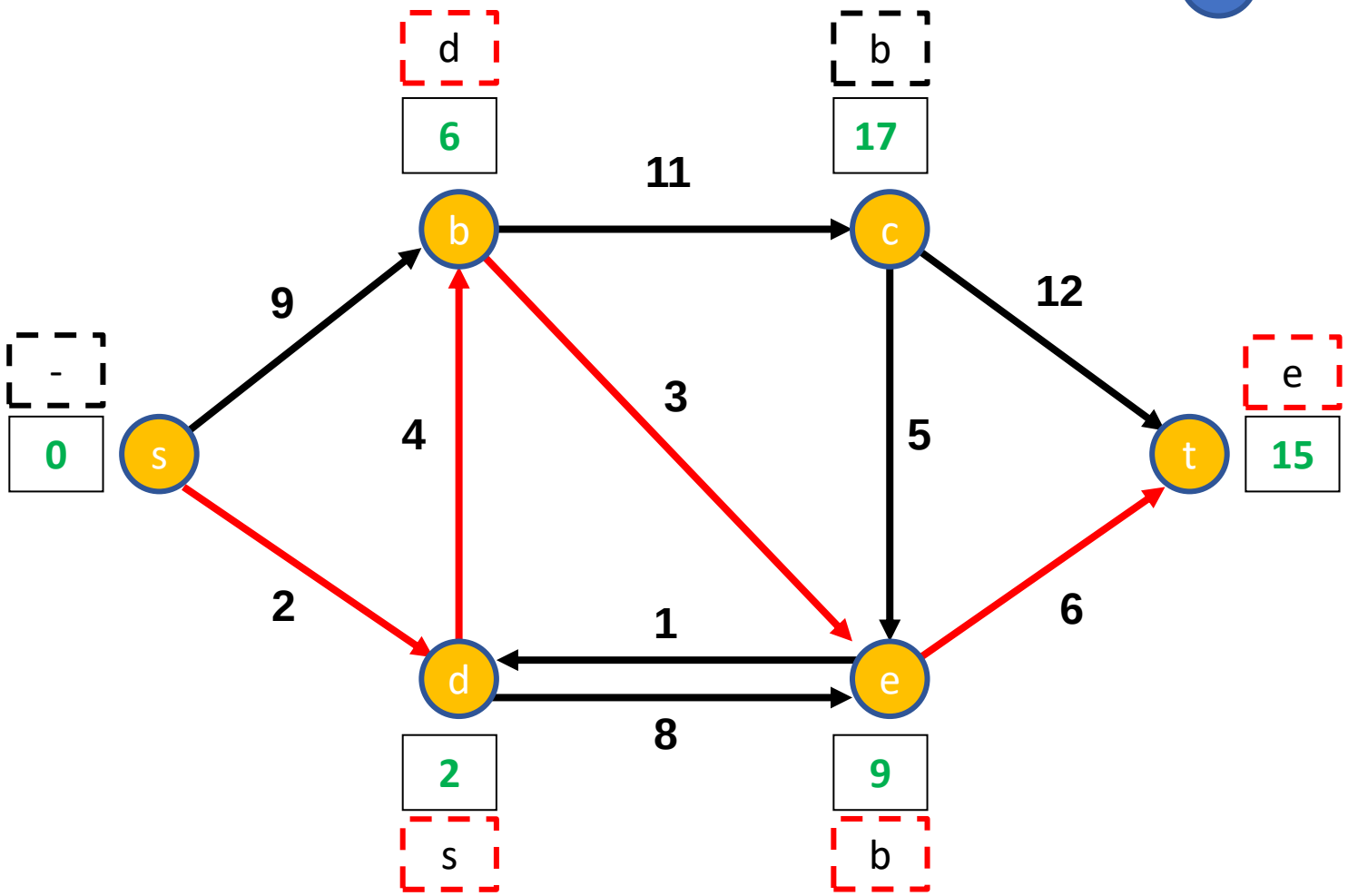


To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e

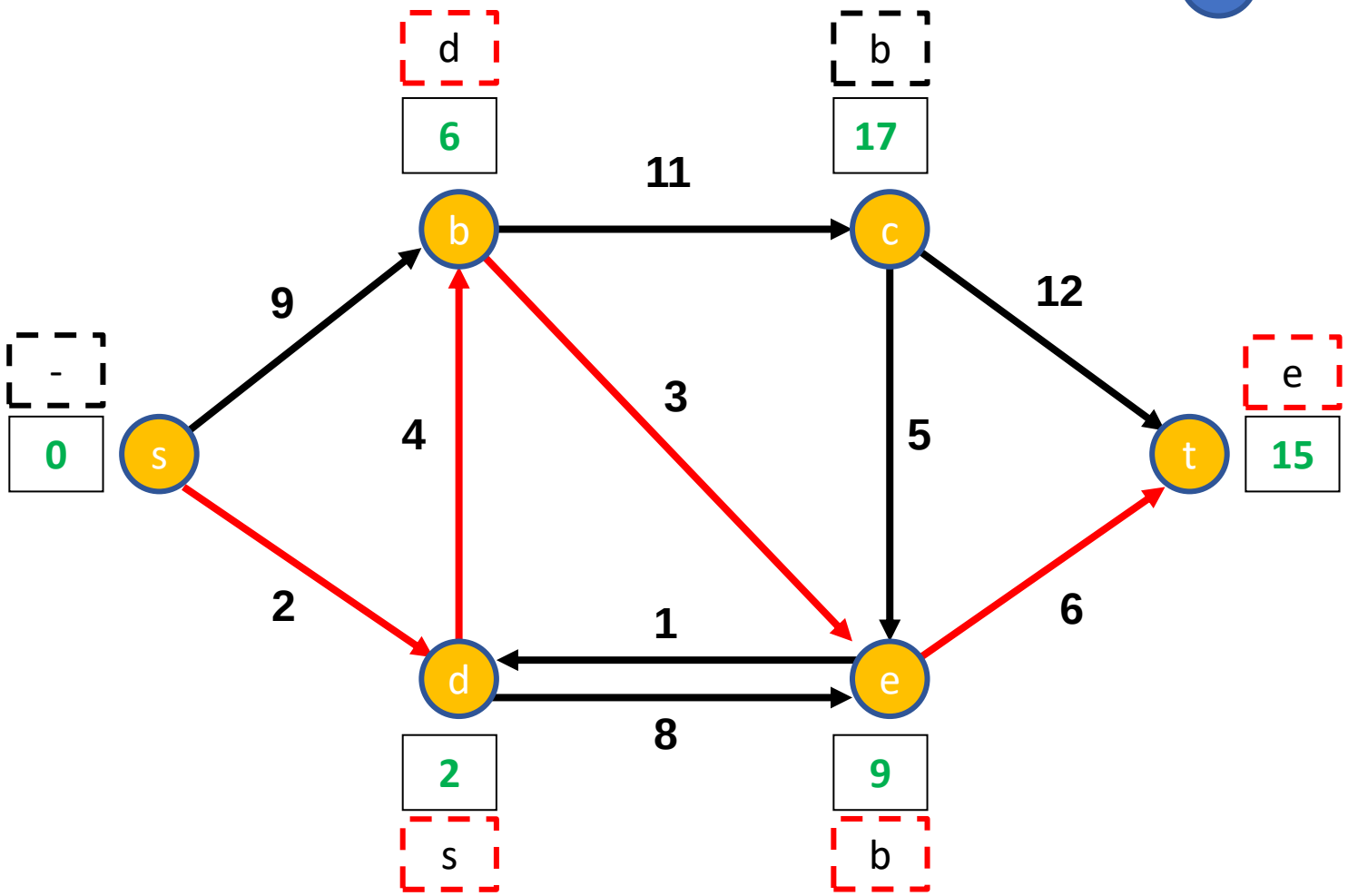


To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

Example

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e



Shortest path from s to t is $s - d - b - e - t$ with a path length of 15

Sensitivity: Internal

Correctness

Visits all vertices in increasing order of distance and visits each vertex exactly once.

Dijkstra maintains a set of visited vertices (permanent) and unvisited vertices (temporary).

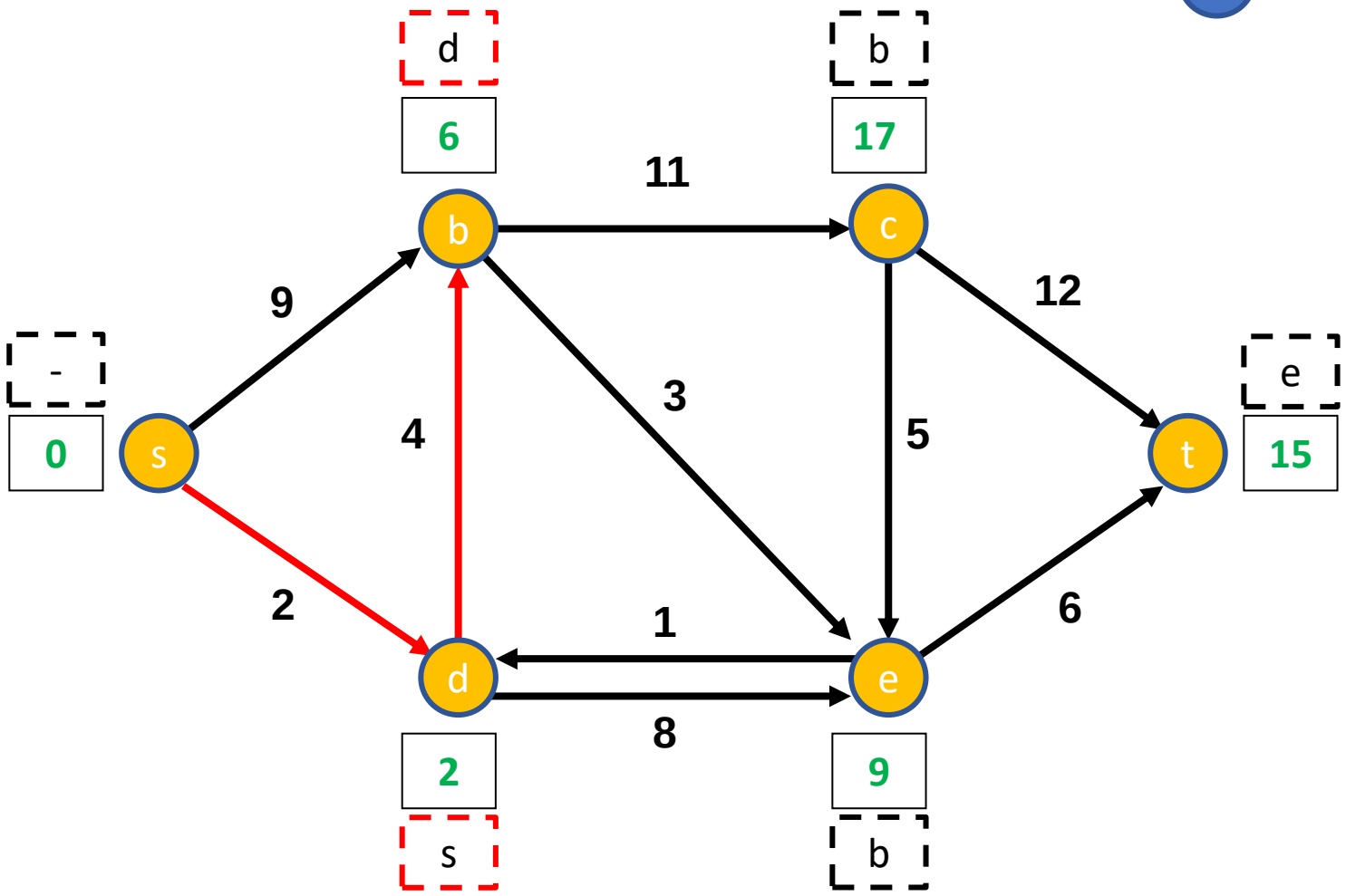
Claim: For a visited vertex v, d_v and π_v will be the optimal shortest path distance and corresponding predecessor vertex.

Meaning, we get all the shortest paths from the source to all visited vertices free!

Sensitivity: Internal

Shortest sub-path

v	Label	d_v	π_v
s	P	0	-
b	P	~~9~~ 6	~~s~~ d
c	P	17	b
d	P	2	s
e	P	~~10~~ 9	~~d~~ b
t	P	15	e



For example the shortest path from vertex s to b is $s - d - b$ of length 6

Sensitivity: Internal

Proof and Time Complexity

Dijkstra's Algorithm has a worst-case time complexity of $O(V^2)$

- $O(|V|^2)$ with a simple array (scan every vertex for the minimum)
- $O((|E| + |V|) \log |V|)$ with an adjacency list and a binary heap

If you are interested you can find all the proofs about the correctness and time complexities of Dijkstra's and the other algorithms in:

Cormen, T. H., Leiserson, C. E., Rivest, R. L., Stein, C. (2001). *Introduction to Algorithms*. The MIT Press. ISBN: 0262032937

Or check out MIT 6.006 Lecture 16 (Dijkstra) in this week's 6 Further Reading folder.

Note they are somewhat complicated if you haven't seen this type of argument before.

Sensitivity: Internal

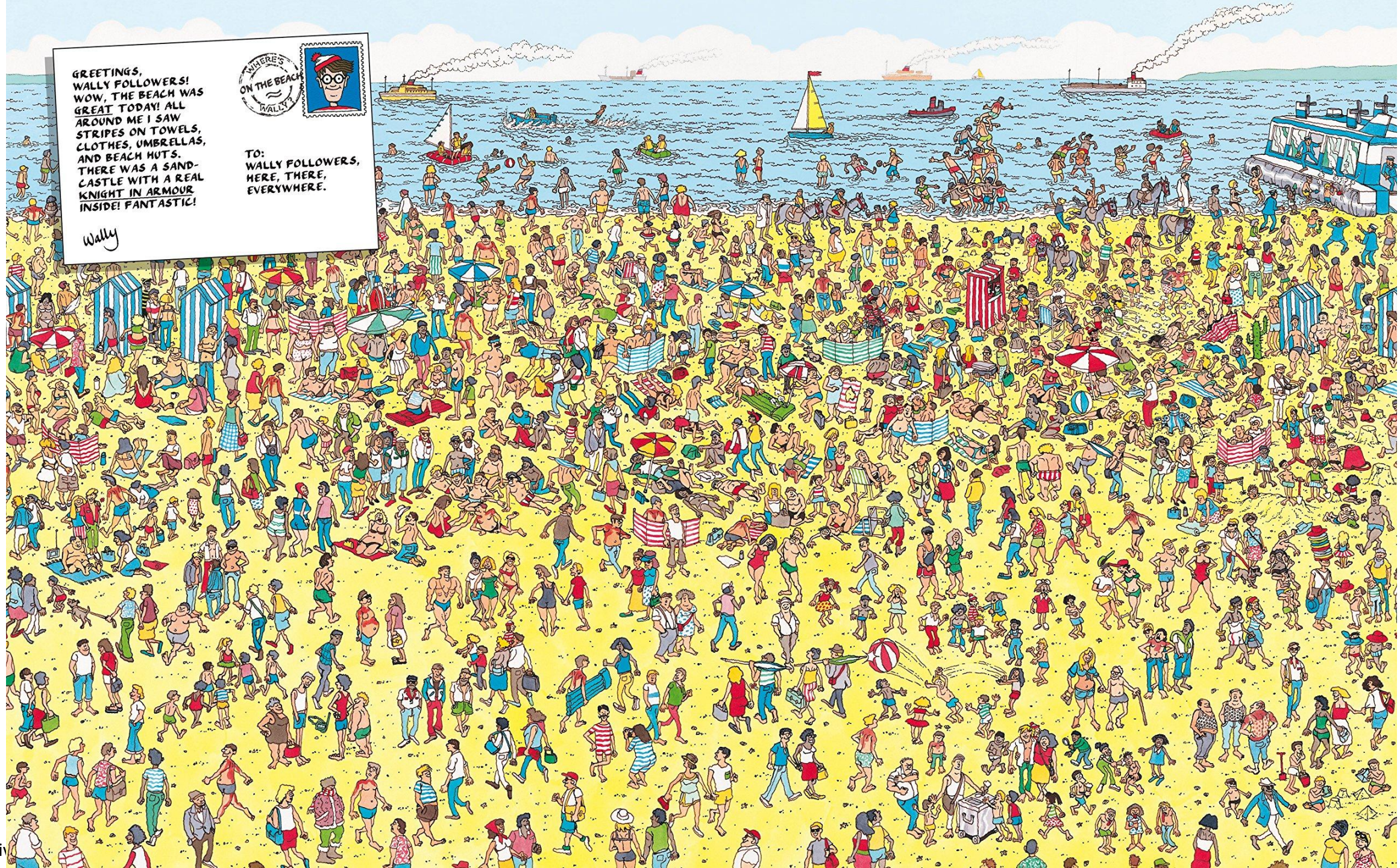
30 seconds....

You have 30 seconds to find Wally.

Raise your hand if you find him and I will pause to check.



Sensitivity: Internal

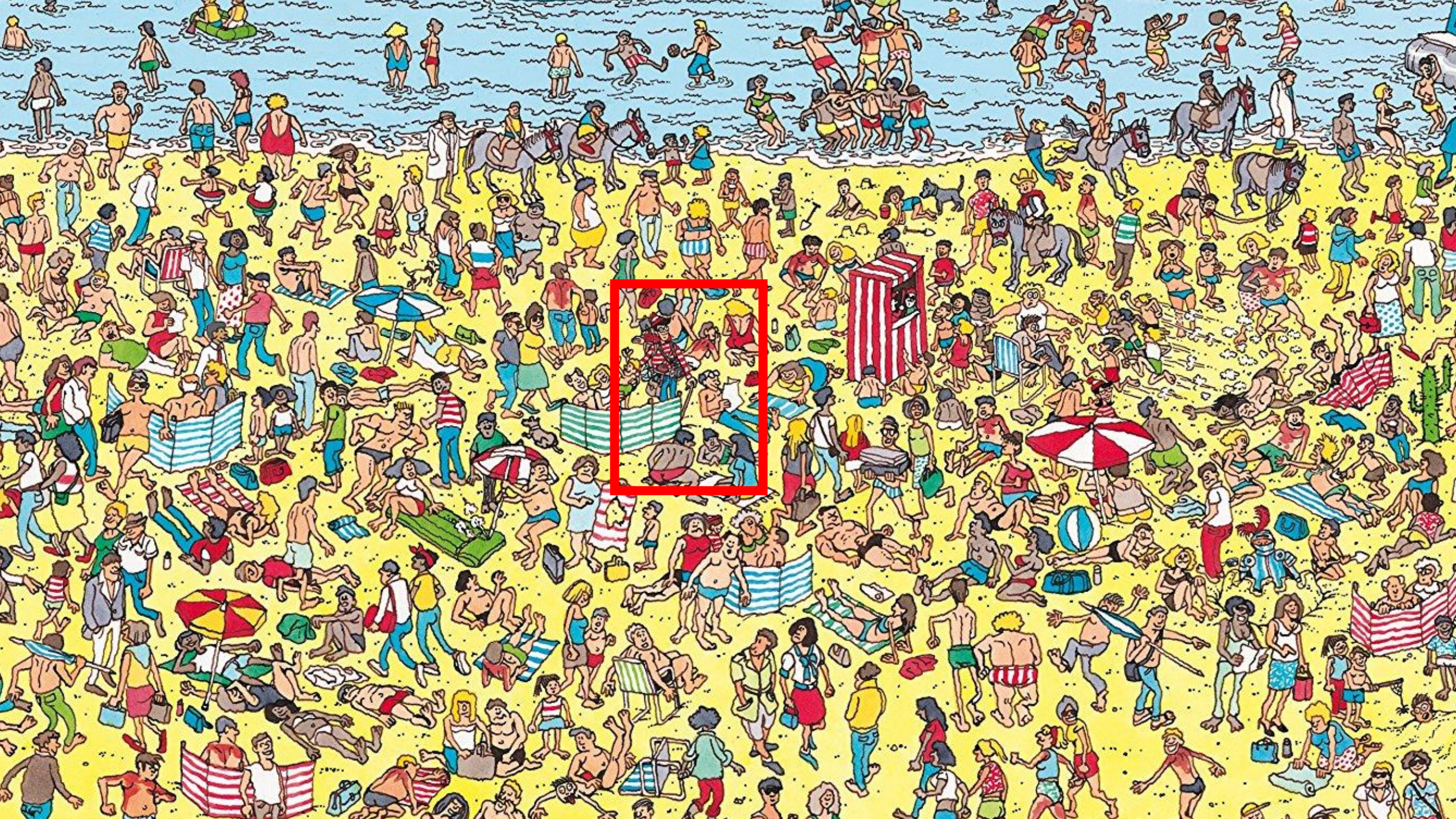

GREETINGS,
WALLY FOLLOWERS!
WOW, THE BEACH WAS
GREAT TODAY! ALL
AROUND ME I SAW
STRIPES ON TOWELS,
CLOTHES, UMBRELLAS,
AND BEACH HUTS.
THERE WAS A SAND-
CASTLE WITH A REAL
KNIGHT IN ARMOUR
INSIDE! FANTASTIC!
Wally
WHERE'S WALLY?
ON THE BEACH
TO:
WALLY FOLLOWERS,
HERE, THERE,
EVERYWHERE.

Sensitiv

Checking Answer….

Please be patient, there is some high-tech stuff going on right now…

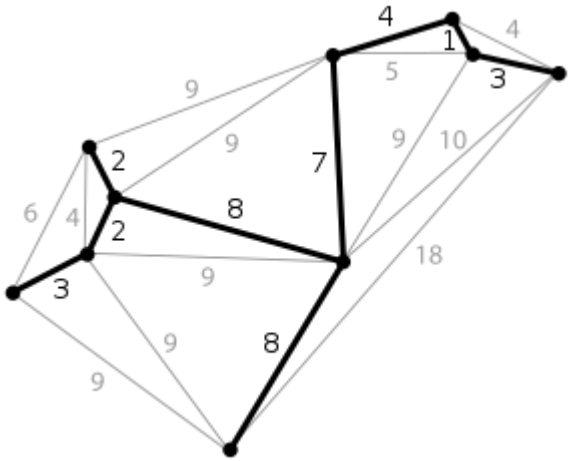
Sensitivity: Internal



Minimum Spanning Tree

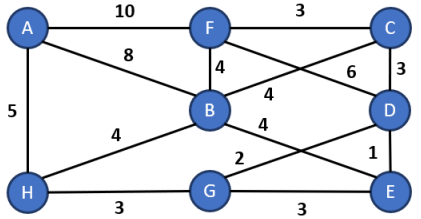
Definition

A subset of weighted edges that connects all the vertices of a connected, weighted, undirected graph together without any cycles and has the minimum total edge weight.



Minimum Spanning Tree

Kruskal's Algorithm



Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	
(D,G)	2	
(E,G)	3	
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Example Application

A fibre optics company is looking to expand its portfolio by providing broadband to an entirely new set of areas. Currently it has cable laid to a particular location.

The cable is extremely expensive and particularly difficult to lay in certain areas, therefore the company wishes to minimise the cost of laying the cable, whilst connecting all areas.

The company can lay cable between certain areas and has approximated the cost. Compute the minimum cost of connecting all the areas.

Sensitivity: Internal

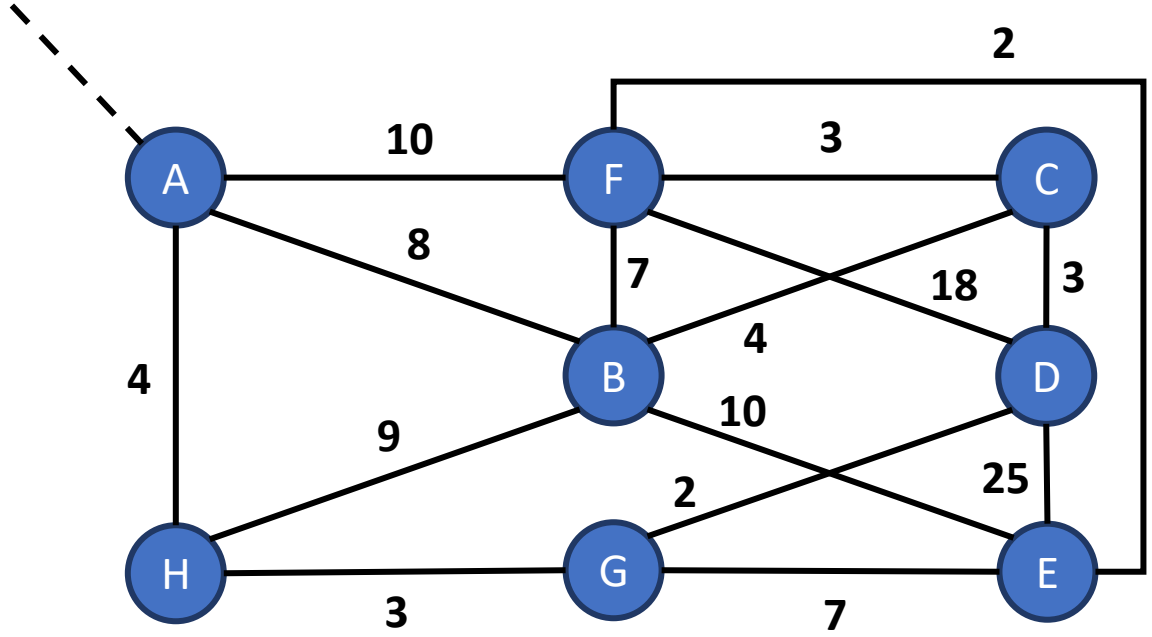
Prim’s Algorithm

1. Initialise distances to vertices as ∞ and predecessors as $-$
2. Arbitrarily select a vertex and label as selected
3. Update distances and predecessors to adjacent unselected vertices
4. Select the reachable vertex with minimum distance
5. Repeat 3 and 4 until all vertices have been selected

Sensitivity: Internal

Prim's Algorithm

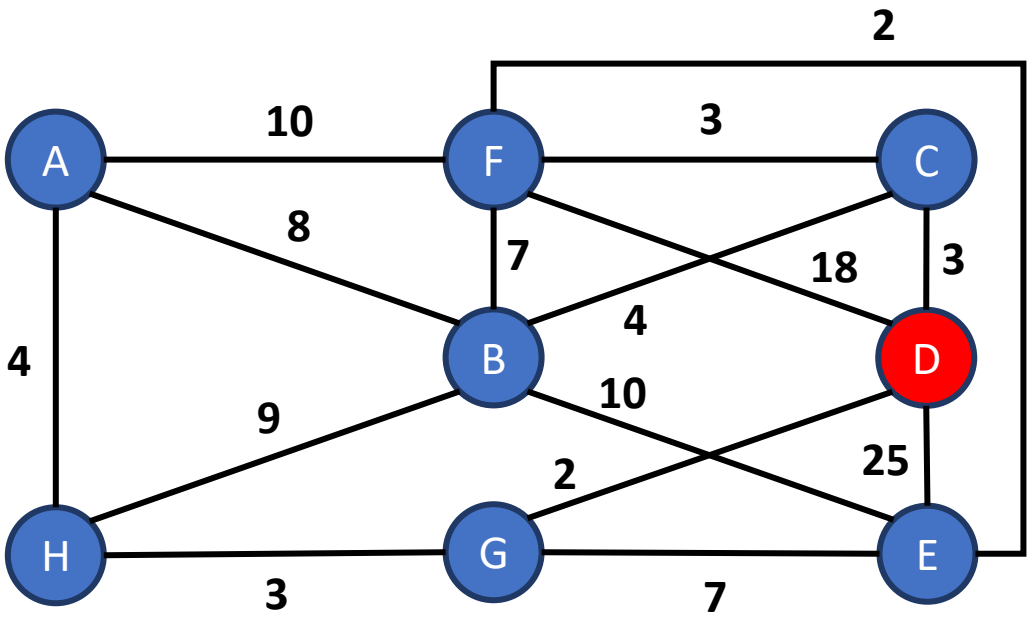
Existing Cable



v	Selected	d_v	π_v
A			
B			
C			
D			
E			
F			
G			
H			

Sensitivity: Internal

Prim's Algorithm

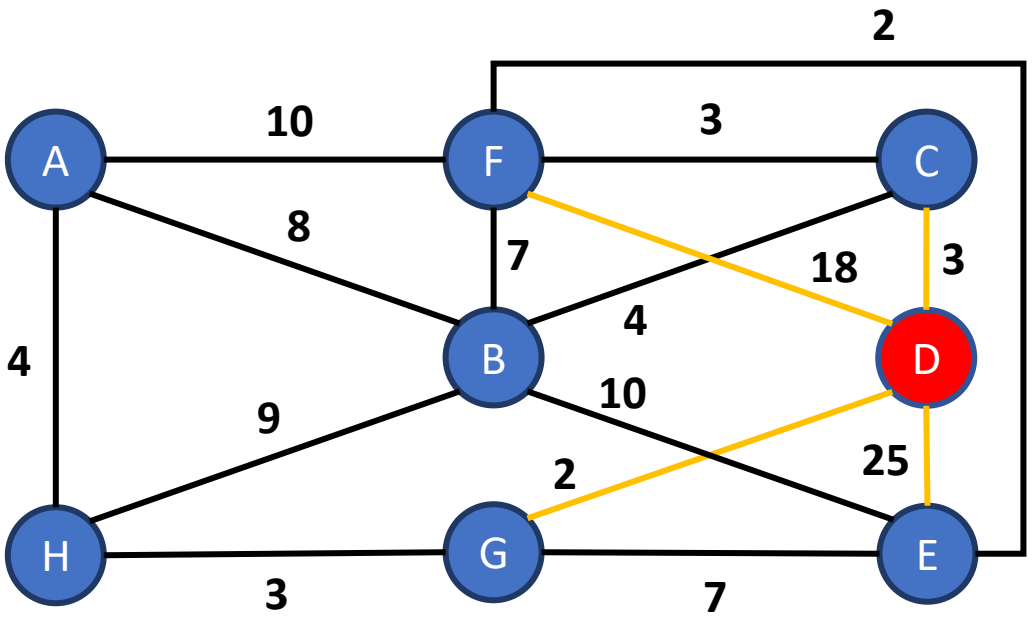


v	Selected	d_v	π_v
A			
B			
C			
D	T	0	-
E			
F			
G			
H			

Arbitrarily select a vertex and mark as selected (here D; any start vertex gives the same total weight).

Sensitivity: Internal

Prim's Algorithm

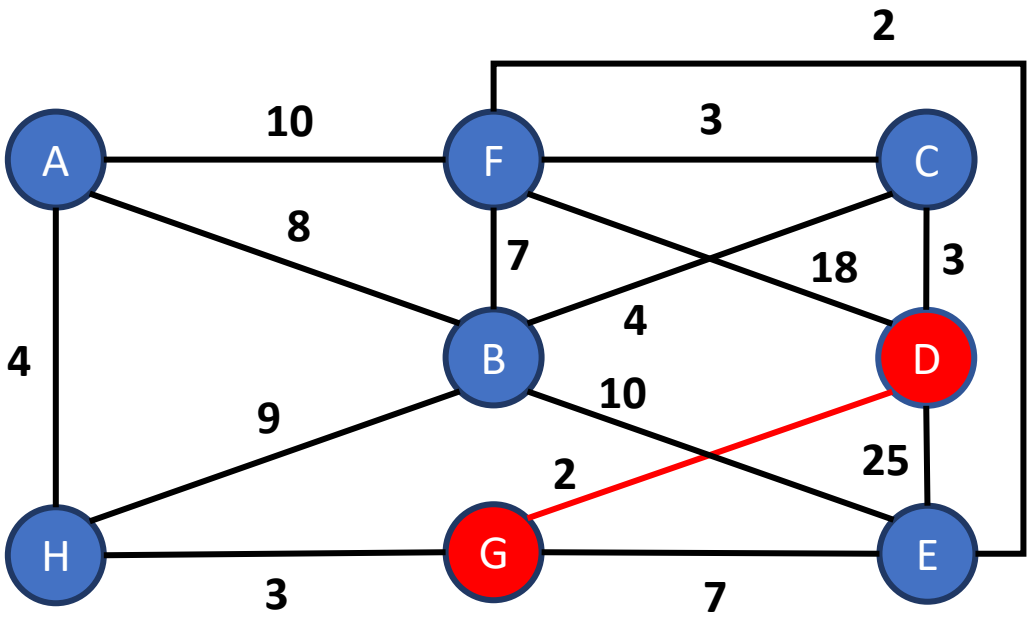


v	Selected	d_v	π_v
A			
B			
C		3	D
D	T	0	-
E		25	D
F		18	D
G		2	D
H			

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

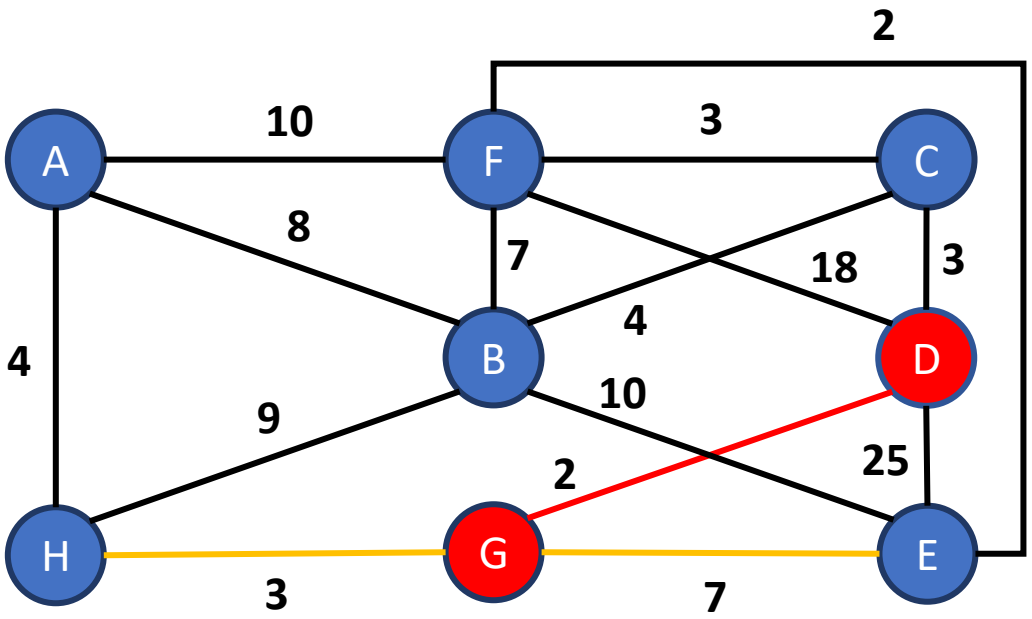


v	Selected	d_v	π_v
A			
B			
C		3	D
D	T	0	-
E		25	D
F		18	D
G	T	2	D
H			

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

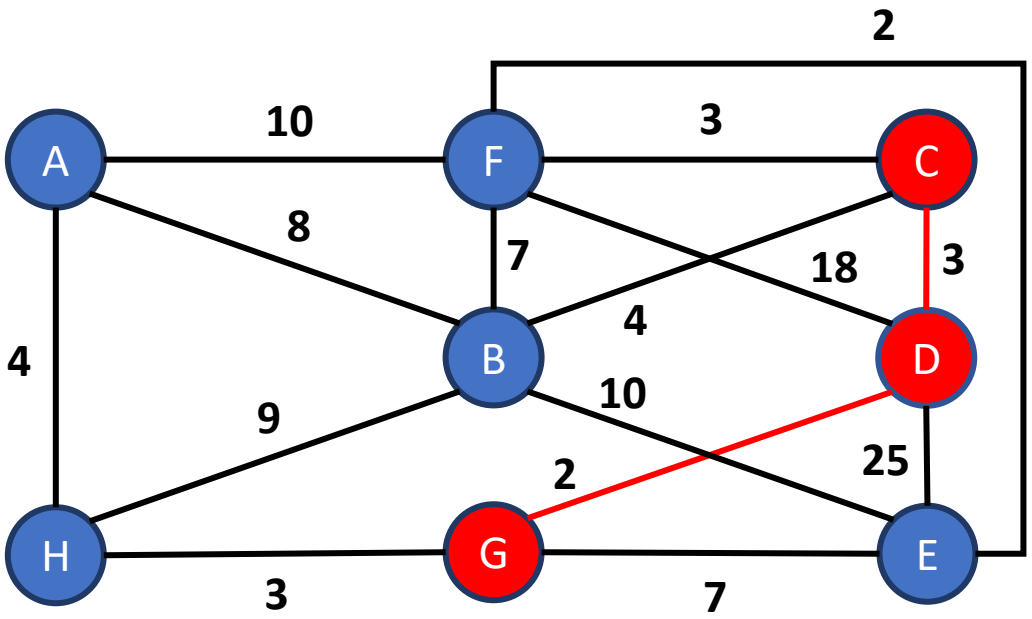


v	Selected	d_v	π_v
A			
B			
C		3	D
D	T	0	-
E		~~25~~ 7	~~D~~ G
F		18	D
G	T	2	D
H		3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

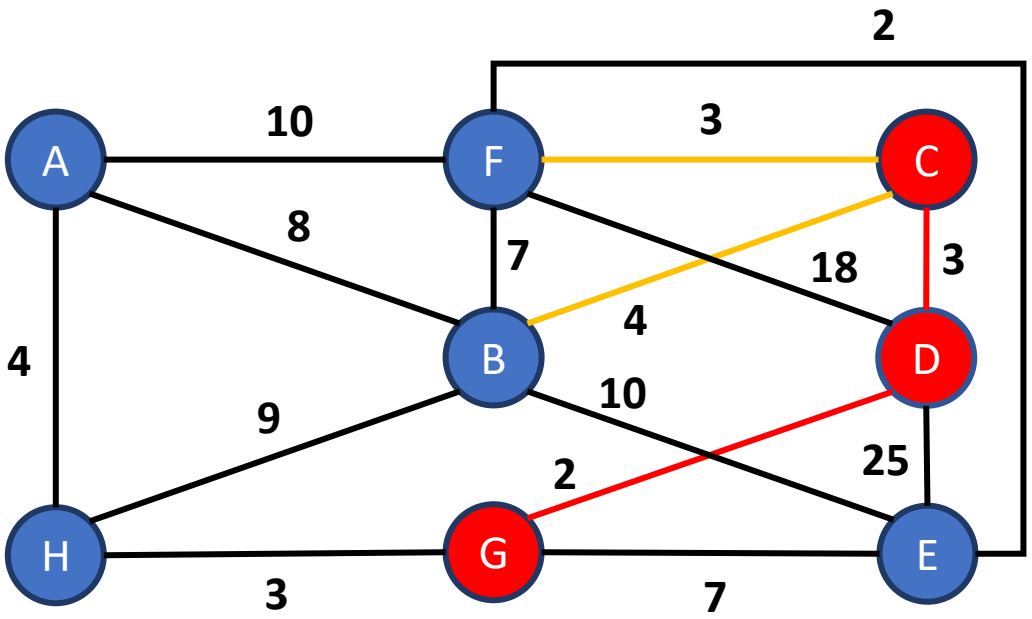


v	Selected	d_v	π_v
A			
B			
C	T	3	D
D	T	0	-
E		~~25~~ 7	~~D~~ G
F		18	D
G	T	2	D
H		3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

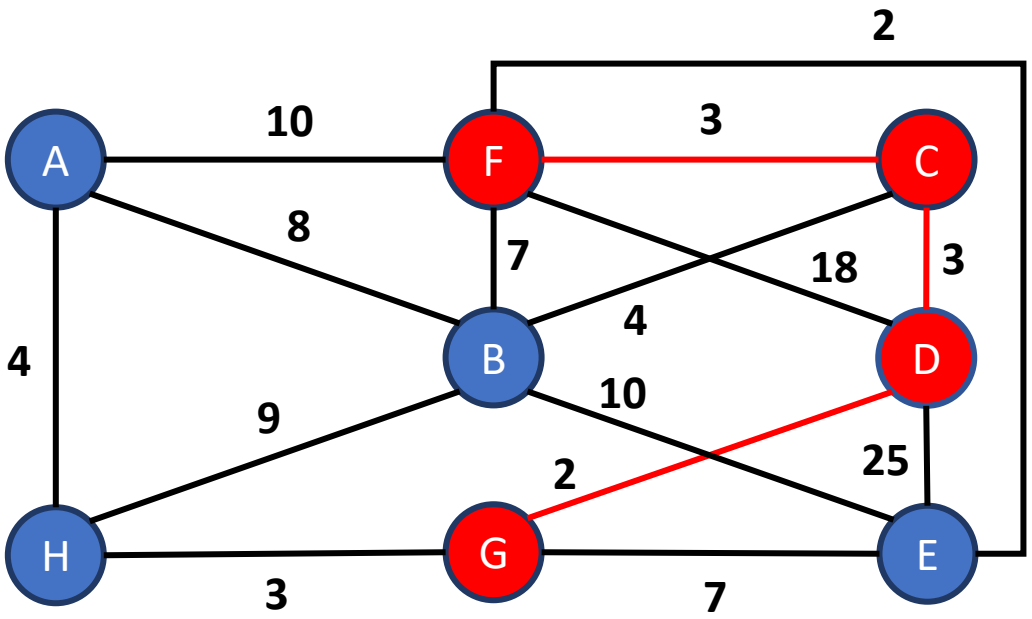


v	Selected	d_v	π_v
A			
B		4	C
C	T	3	D
D	T	0	-
E		~~25~~ 7	G
F		~~18~~ 3	~~D~~ C
G	T	2	D
H		3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

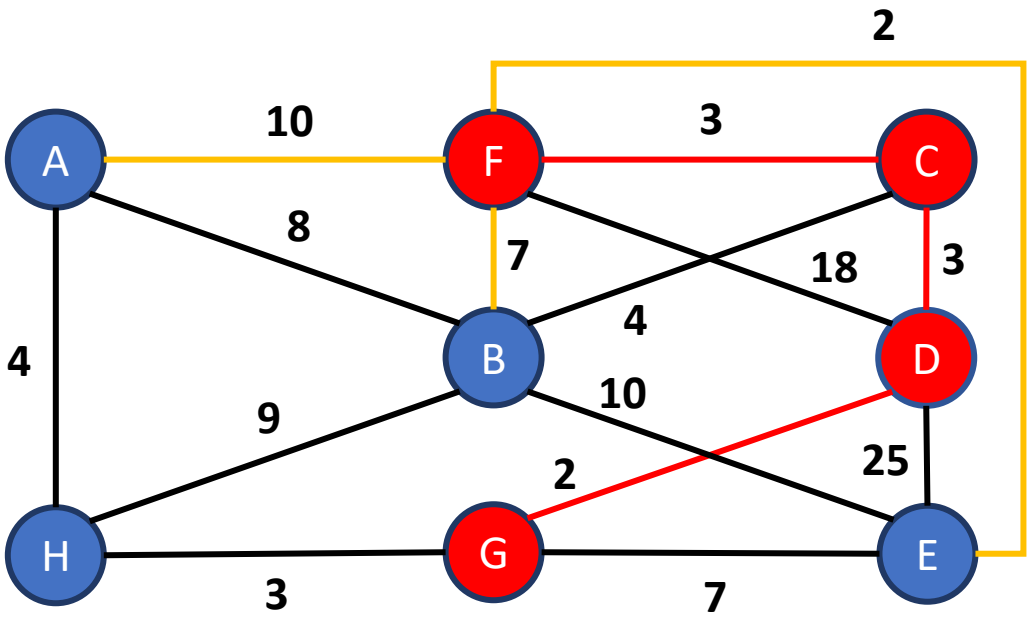


v	Selected	d_v	π_v
A			
B		4	C
C	T	3	D
D	T	0	-
E		~~25~~ 7	G
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H		3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

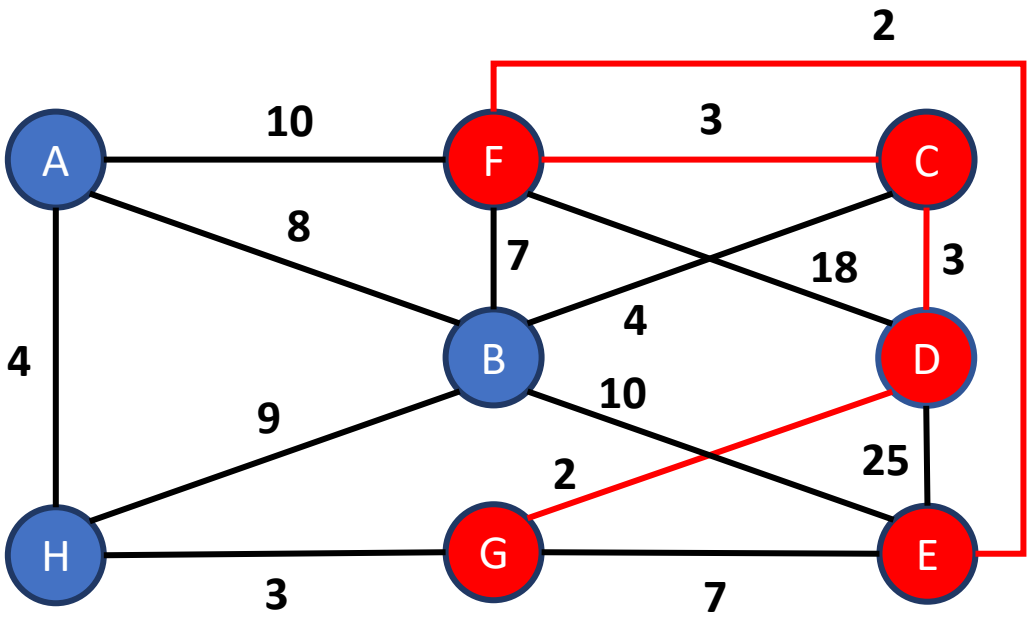


v	Selected	d_v	π_v
A		10	F
B		4	C
C	T	3	D
D	T	0	-
E		~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H		3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

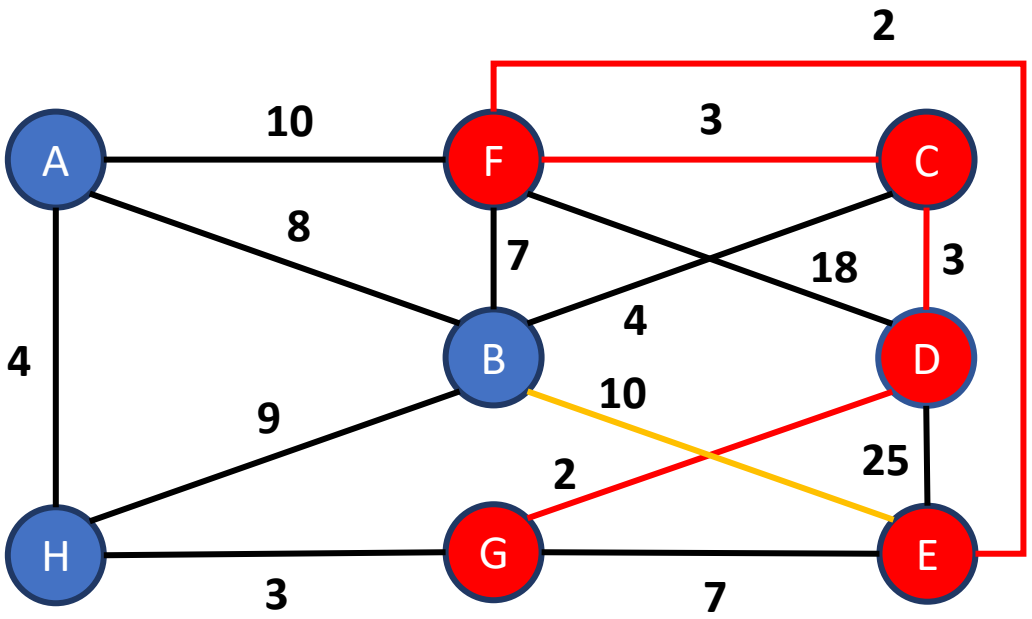


v	Selected	d_v	π_v
A		10	F
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H		3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

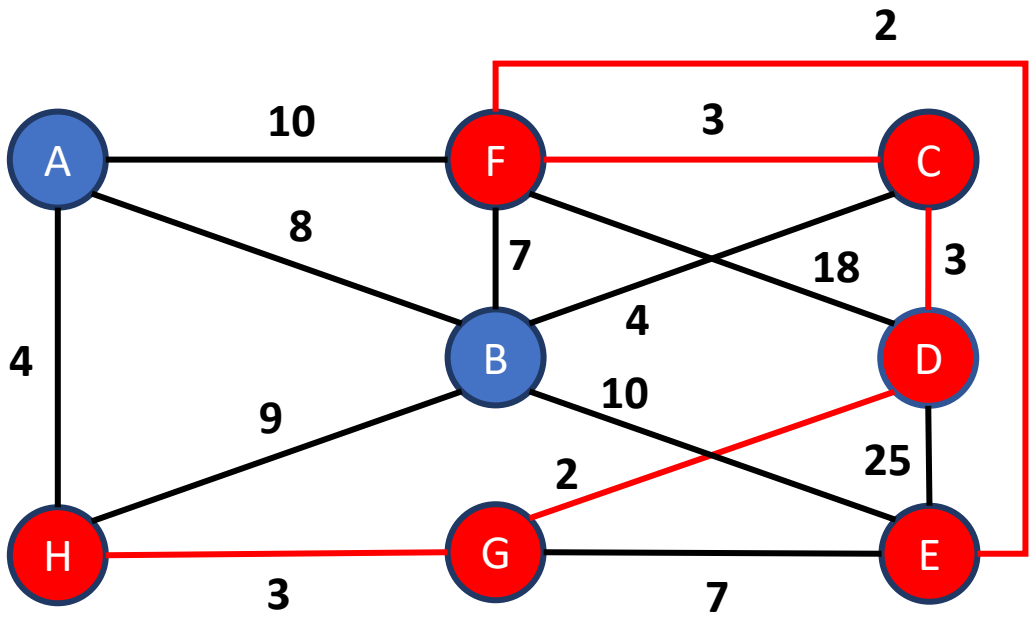


v	Selected	d_v	π_v
A		10	F
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H		3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

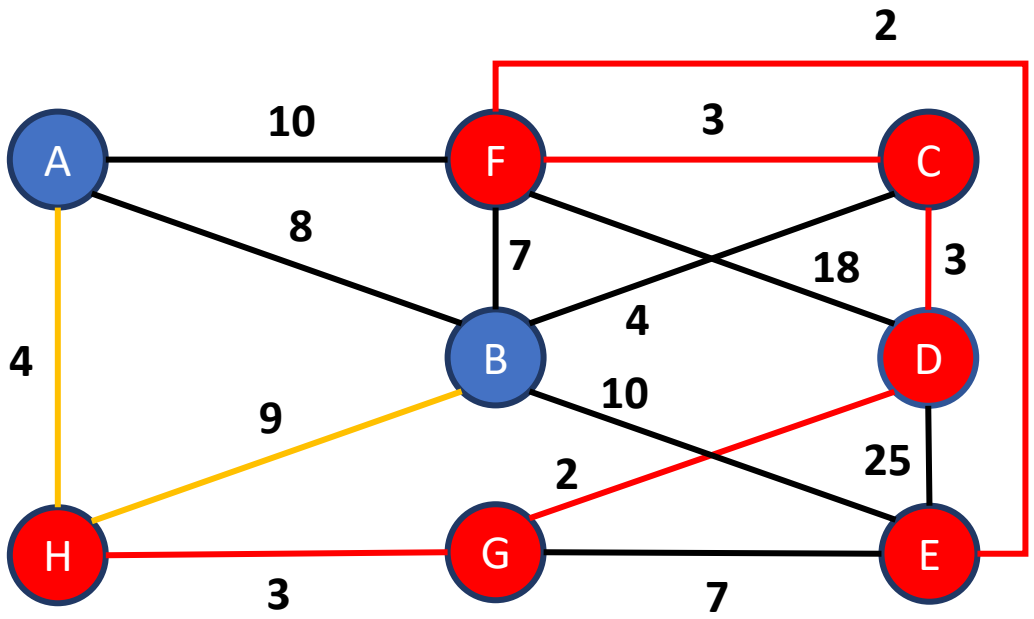


v	Selected	d_v	π_v
A		10	F
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

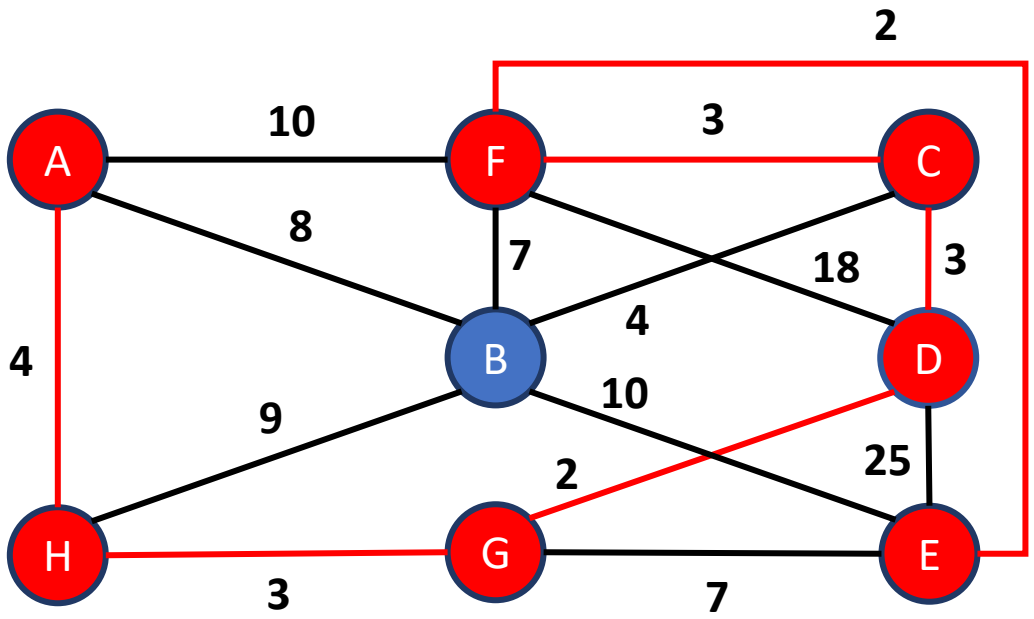


v	Selected	d_v	π_v
A		~~10~~ 4	~~F~~ H
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

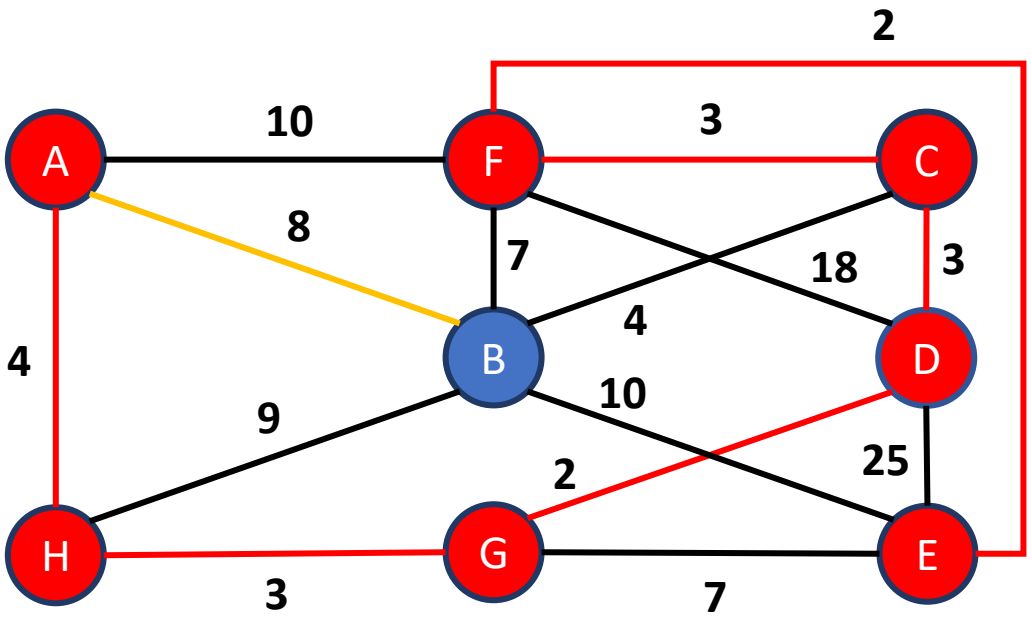


v	Selected	d_v	π_v
A	T	~~10~~ 4	~~F~~ H
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm

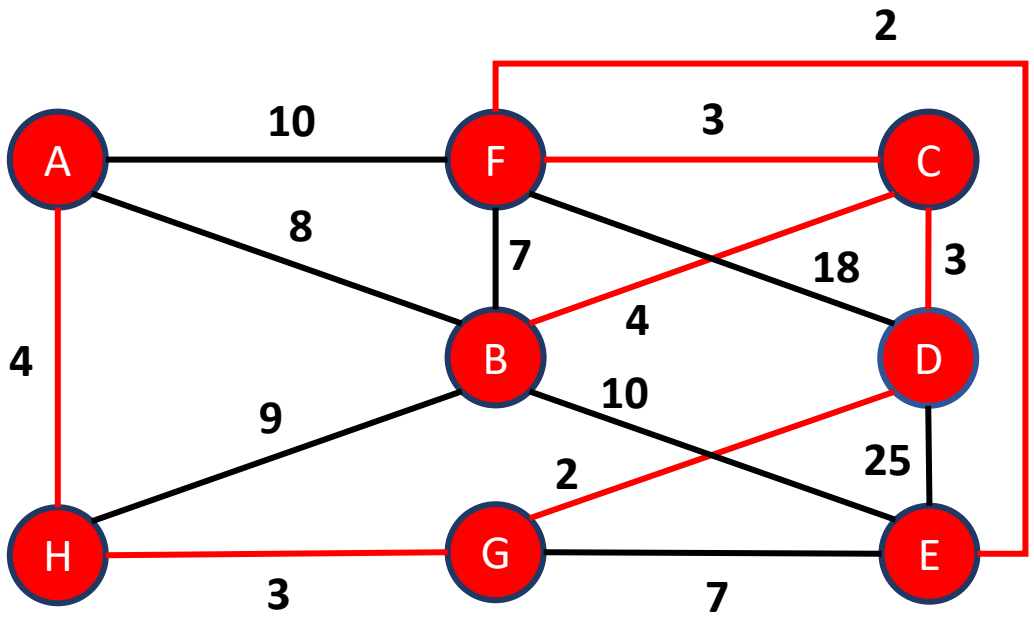


v	Selected	d_v	π_v
A	T	~~10~~ 4	~~F~~ H
B		4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

Update distances of adjacent, unselected vertices

Sensitivity: Internal

Prim's Algorithm

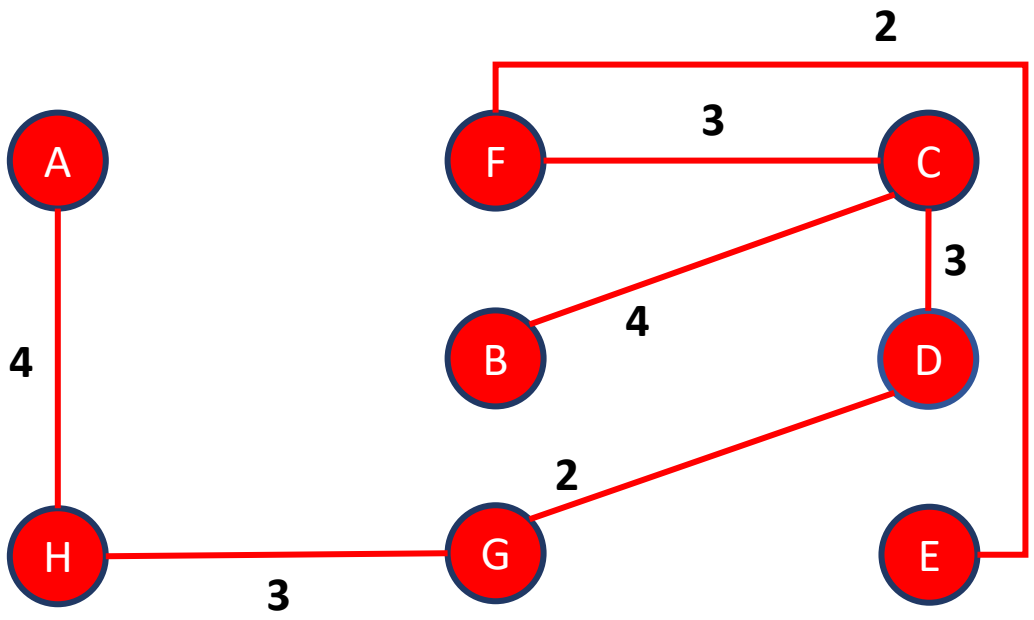


v	Selected	d_v	π_v
A	T	~~10~~ 4	~~F~~ H
B	T	4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

Select reachable vertex with minimum distance

Sensitivity: Internal

Prim's Algorithm



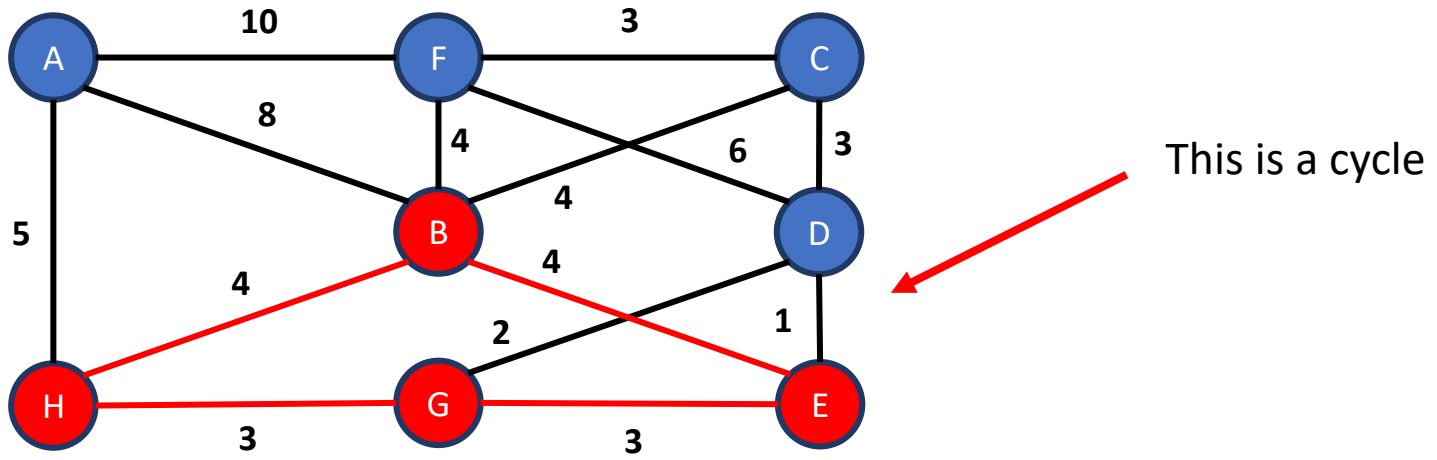
v	Selected	d_v	π_v
A	T	~~10~~ 4	~~F~~ H
B	T	4	C
C	T	3	D
D	T	0	-
E	T	~~25~~ ~~7~~ 2	~~G~~ F
F	T	~~18~~ 3	~~D~~ C
G	T	2	D
H	T	3	G

All vertices added and we have the minimum spanning tree. $\sum_v d_v = 21$

Sensitivity: Internal

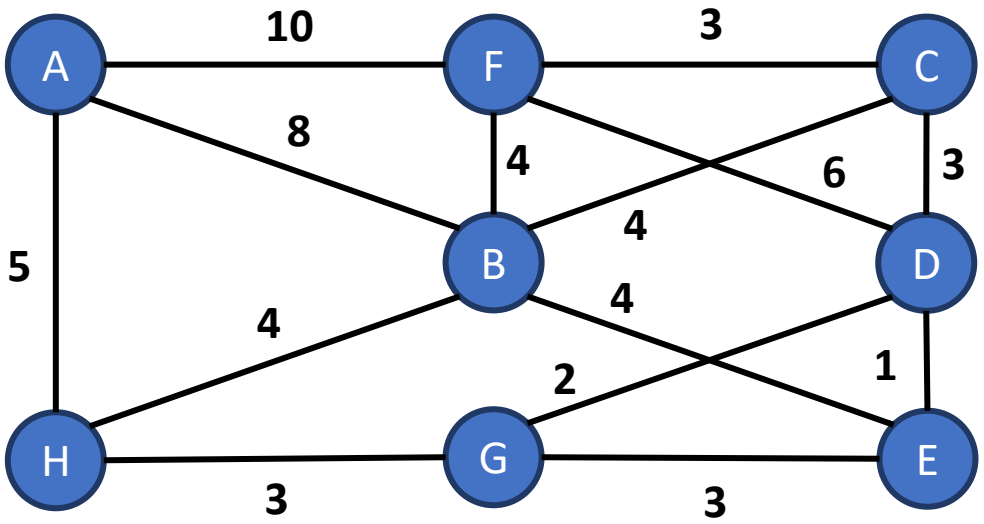
Kruskal's Algorithm

1. Order the edges by weightings, lowest to highest
2. Select the lowest-weight edge not yet considered; if it does not form a cycle, label it as selected
3. Repeat step 2 until all vertices have been selected



Sensitivity: Internal

Kruskal's Algorithm

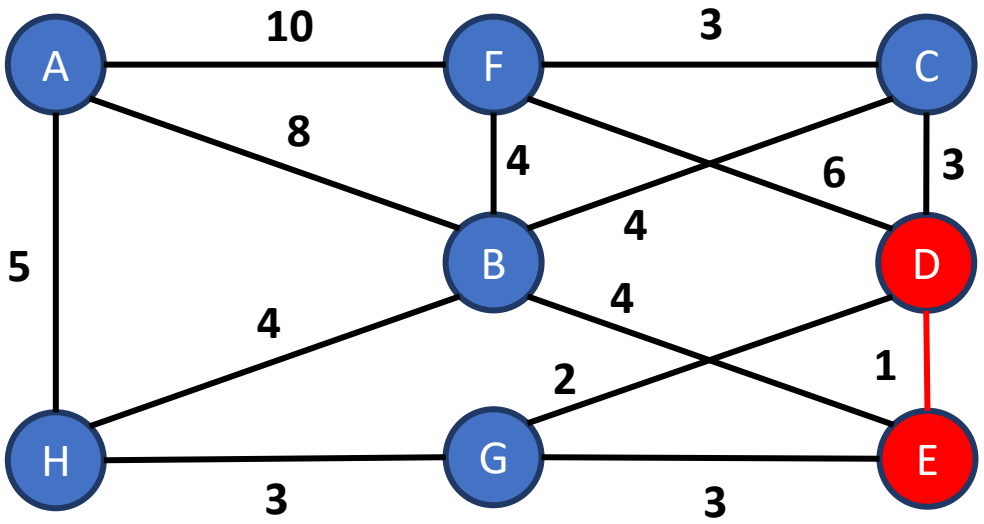


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	
(D,G)	2	
(E,G)	3	
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

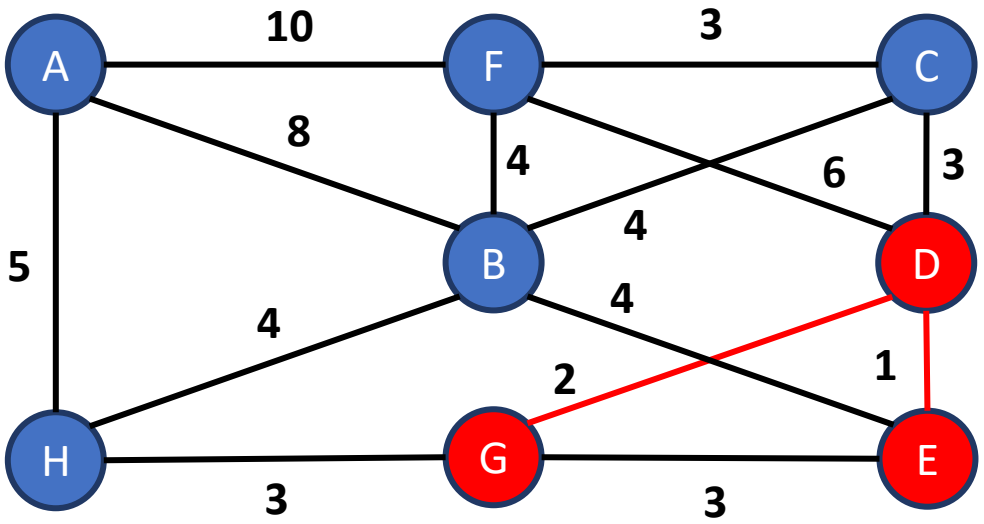


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	
(E,G)	3	
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

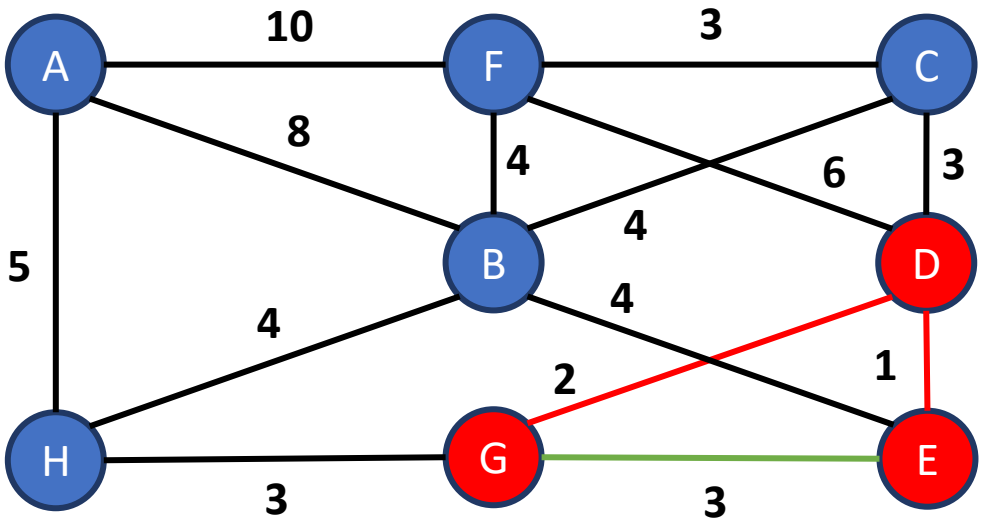


Select first $|V| - 1$ edges that do not form a cycle

Edge	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



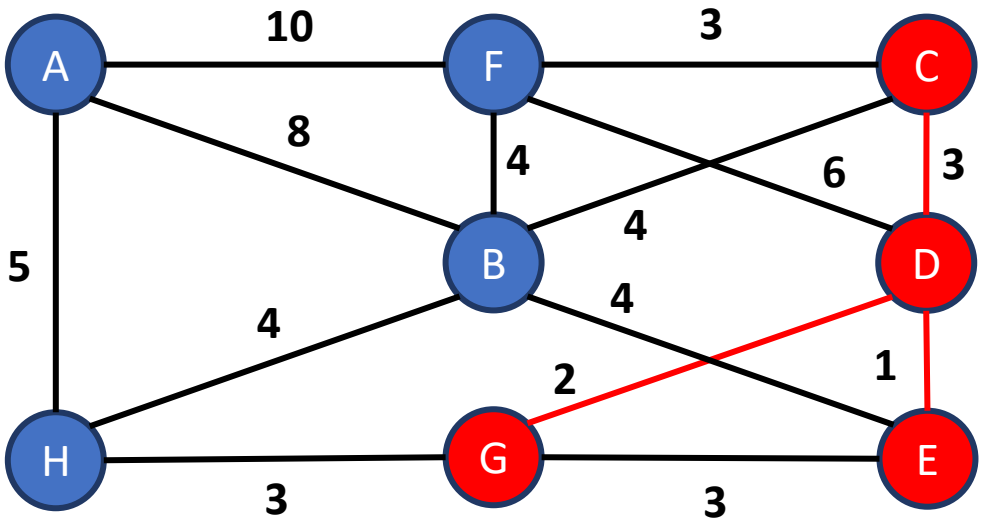
Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

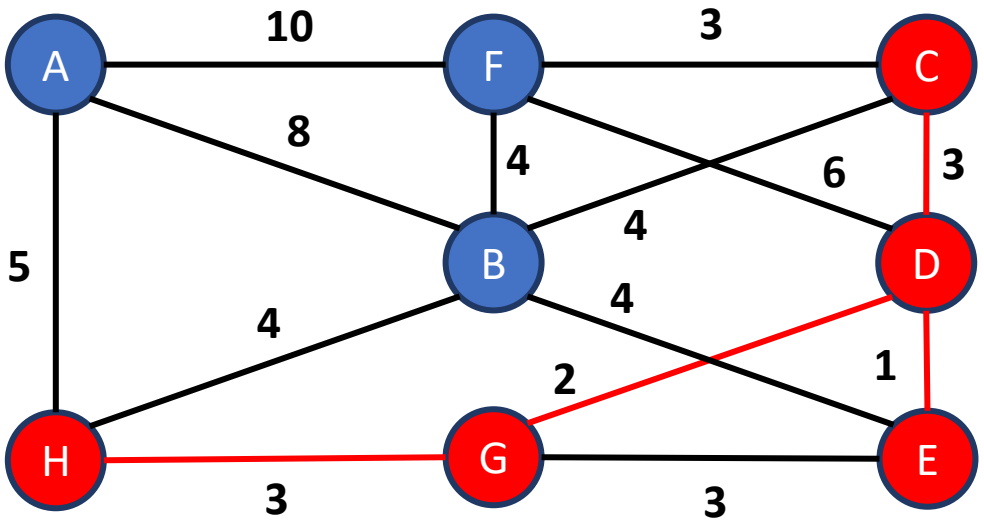


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

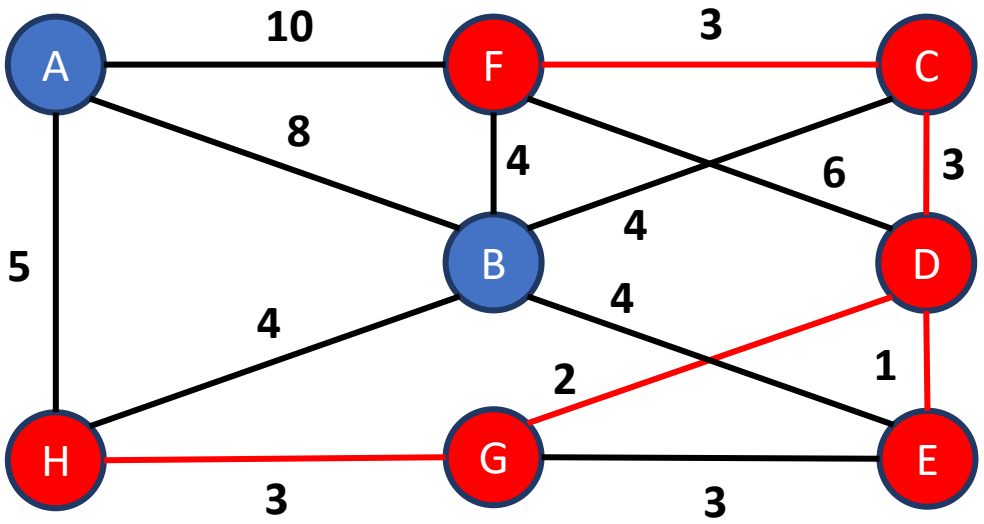


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

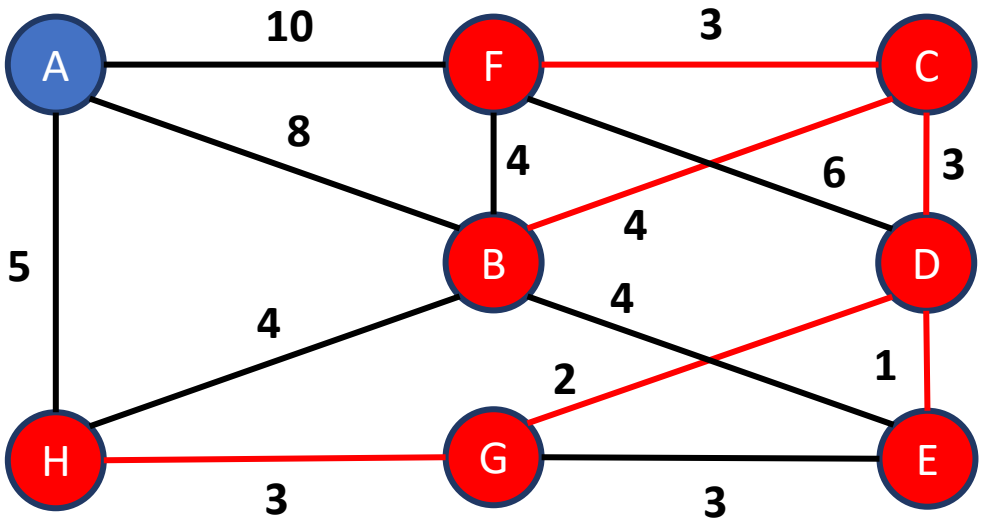


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm

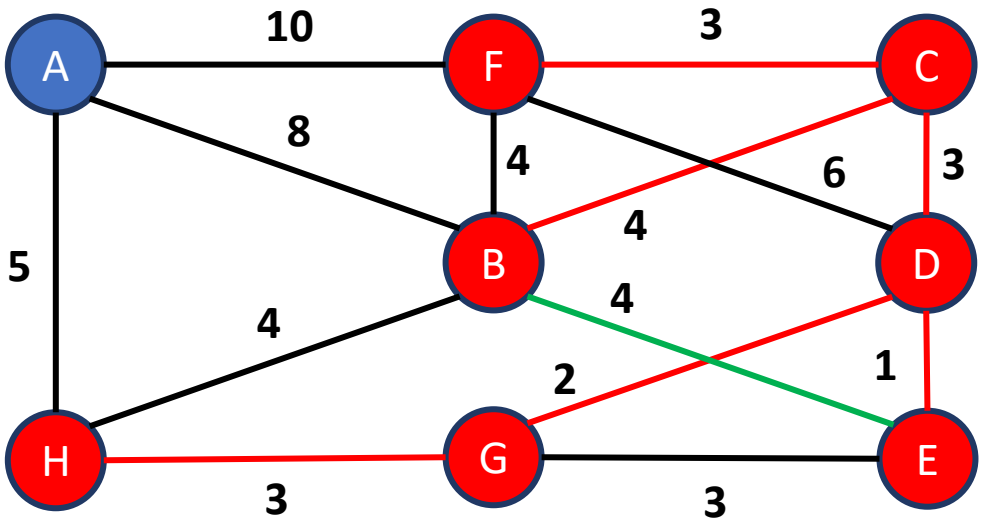


Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	T
(B,E)	4	
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



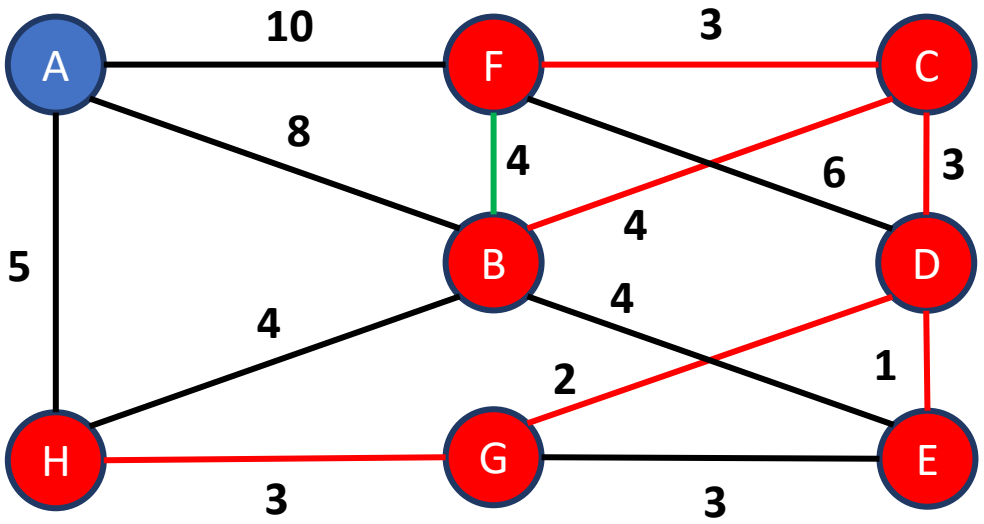
Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	T
(B,E)	4	F
(B,F)	4	
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



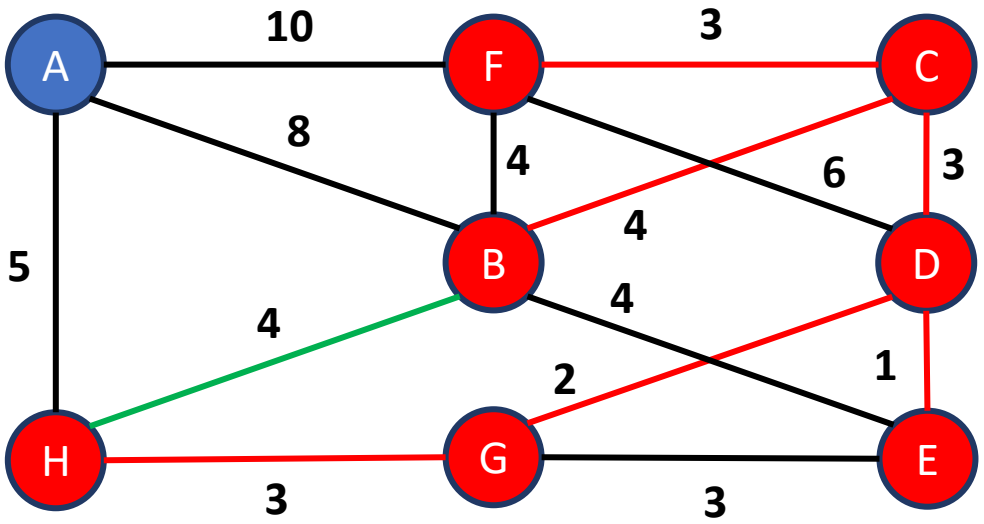
Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	T
(B,E)	4	F
(B,F)	4	F
(B,H)	4	
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



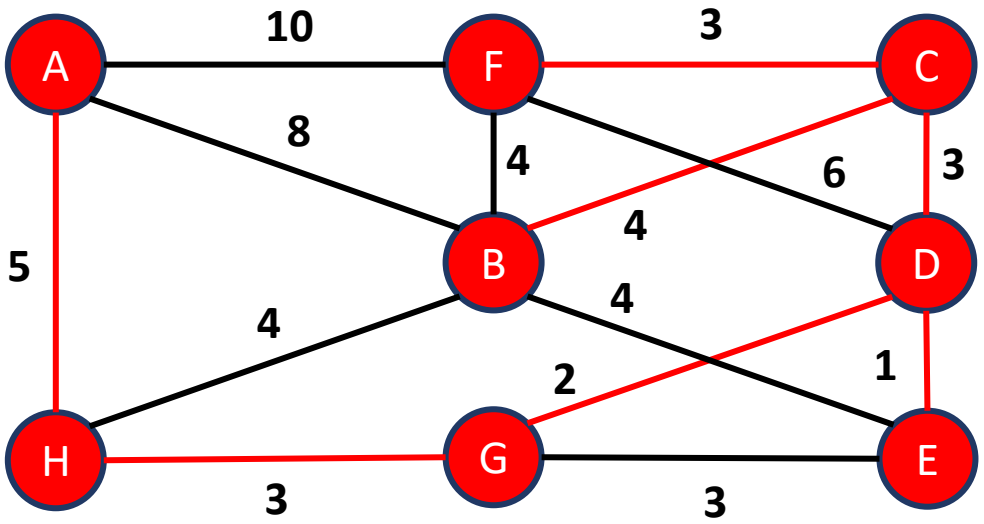
Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

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(D,E)	1	T
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(C,F)	3	T
(B,C)	4	T
(B,E)	4	F
(B,F)	4	F
(B,H)	4	F
(A,H)	5	
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



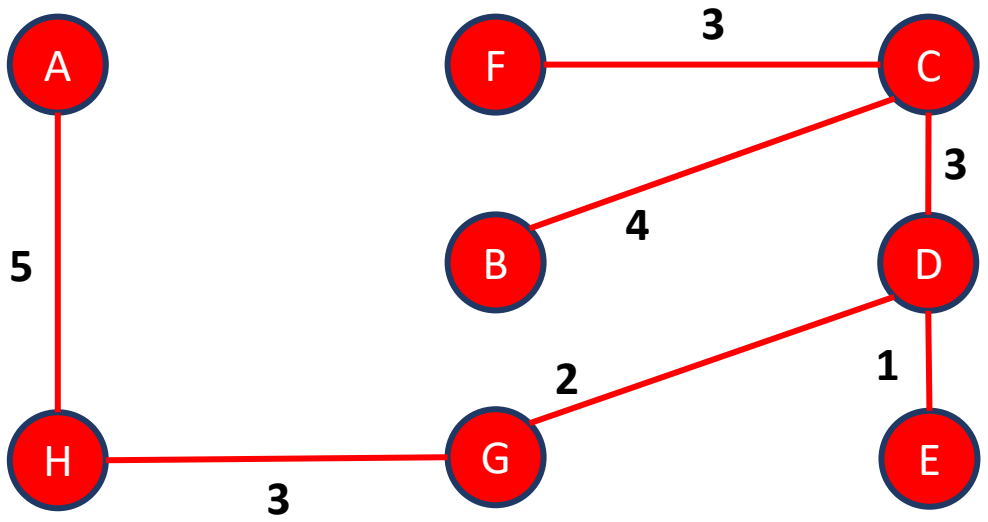
All vertices are selected. Stop

Select first $|V| - 1$ edges that do not form a cycle

Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	T
(B,E)	4	F
(B,F)	4	F
(B,H)	4	F
(A,H)	5	T
(D,F)	6	
(A,B)	8	
(A,F)	10	

Sensitivity: Internal

Kruskal's Algorithm



Edge e	w_e	Selected
(D,E)	1	T
(D,G)	2	T
(E,G)	3	F
(C,D)	3	T
(G,H)	3	T
(C,F)	3	T
(B,C)	4	T
(B,E)	4	F
(B,F)	4	F
(B,H)	4	F
(A,H)	5	T
(D,F)	6	F
(A,B)	8	F
(A,F)	10	F

$|V| - 1$ edges selected, let A be the edges in the MST, then

$$\sum_{e \in A} w_e = 21$$

Sensitivity: Internal

References

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Sensitivity: Internal